CB008247 SRS based downlink beamforming

- CB010602 SRS based Beamforming extension
- CB010388 Support of 5GC001086 and CB008247 using 4DL4UL celltype and 64TRX RUs
- CNI-92005 No PDSCH scheduling in last symbol of SSB slot to mitigate high BLER

Network Engineering Information

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- Author/Presenter: Krzysztof Wisniowski, Norbert Lukasiewicz
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Feature scope matrix

List of features, subfeatures and CRs considered for this presentation

Feature / CR ID	Considered subfeatures	CFAM version	Comment
CB008247 SRS based downlink beamforming	-A: SRS based beamforming with UE specific CSI-RS -B: SRS based beamforming with cell specific CSI-RS (option 1) -D: feature interaction with cDRX,CA,EIRP control, multiple BWP ~~-E: Feature interaction with MU-MIMO~~ -W: Reserve for performance & Capacity -O: Reserve for optimization	1.7	Subfeature-E nas been descoped according to CNI-82066
CB010602 SRS based Beamforming extension	-A: Beam power balancing for PDSCH and UE specific CSI-RS -B: Performance and capacity; Counters	1.0	

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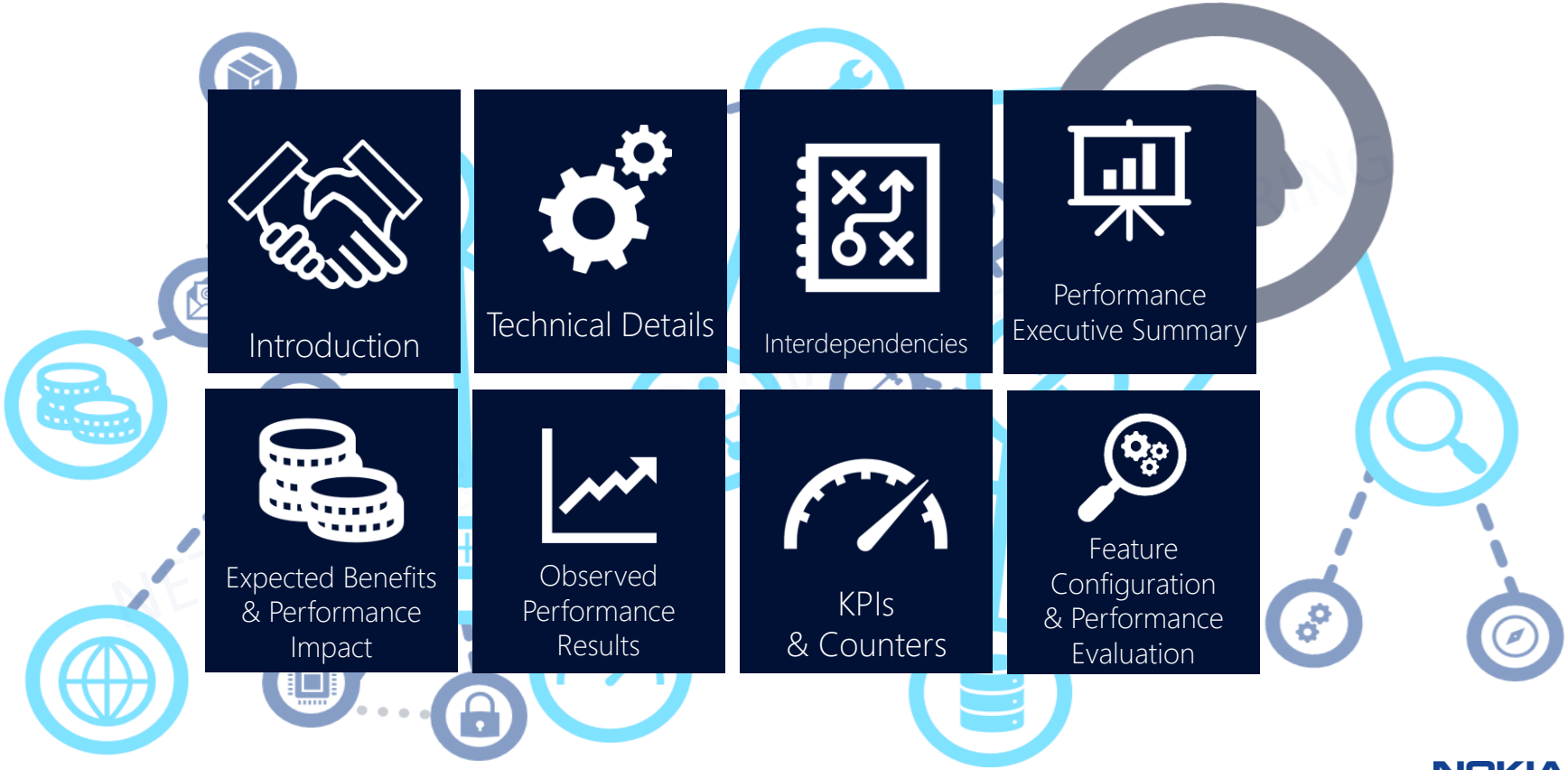
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0.2	C3 ready for review	02-02-2022	K. Wisniowski, N. Lukasiewicz					
0.3	C3 internal review comments considered	14-02-2022	K. Wisniowski, N. Lukasiewicz					
1.0	C3 external review comments considered	22-02-2022	K. Wisniowski, N. Lukasiewicz					
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1.2	Comments received and incorporated	30.03.2022	K. Wisniowski, N. Lukasiewicz					
1.3	CSI-RS co-existnece information added	24-05-2022	K. Wisniowski, N. Lukasiewicz					
1.4	NEI2+ version	24-06-2022	K. Wisniowski, N. Lukasiewicz					
1.5	cSRS/bSRS NIDD parameter recommendations update	01-07-2022	K. Wisniowski, N. Lukasiewicz					
1.5.1	Update on 5GC001086 enbling as mandatory pre-requisite	06-07-2022	K. Wisniowski, N. Lukasiewicz					
1.5.2	NIDD srsForBeamMgmt.srsTypeAllocForDlBm addd	07-07-2022	K. Wisniowski, N. Lukasiewicz					
1.6	Major updates with recent findings	30-08-2022	K. Wisniowski, N. Lukasiewicz					
1.6.1	Added CB010602	25-10-2022	Krzysztof Wisniowski					
1.6.2	Added CB010388 Support of 5GC001086 and CB008247 using 4DL4UL celltype and 64TRX RUs	28-10-2020	K. Wisniowski, N. Lukasiewicz					
1.6.3	Clarification on CB0010602 parametrization	16-01-2023	K. Wisniowski					
1.6.4.1	Added CNI-92005	13-03-2023	K. Wisniowski, N. Lukasiewicz					
1.6.4.2	Editorial corrections, added NIDD parameters related to CNI-92005 and left/right DFT beams muting	14-04-2023	K. Wisniowski, N. Lukasiewicz					
1.6.4.3	Additional explanation on left/right 2D DFT beams muting in Feature Design Section	18-04-2023	K. Wisniowski, N. Lukasiewicz					

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CB008247 SRS based downlink beamforming



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Introduction



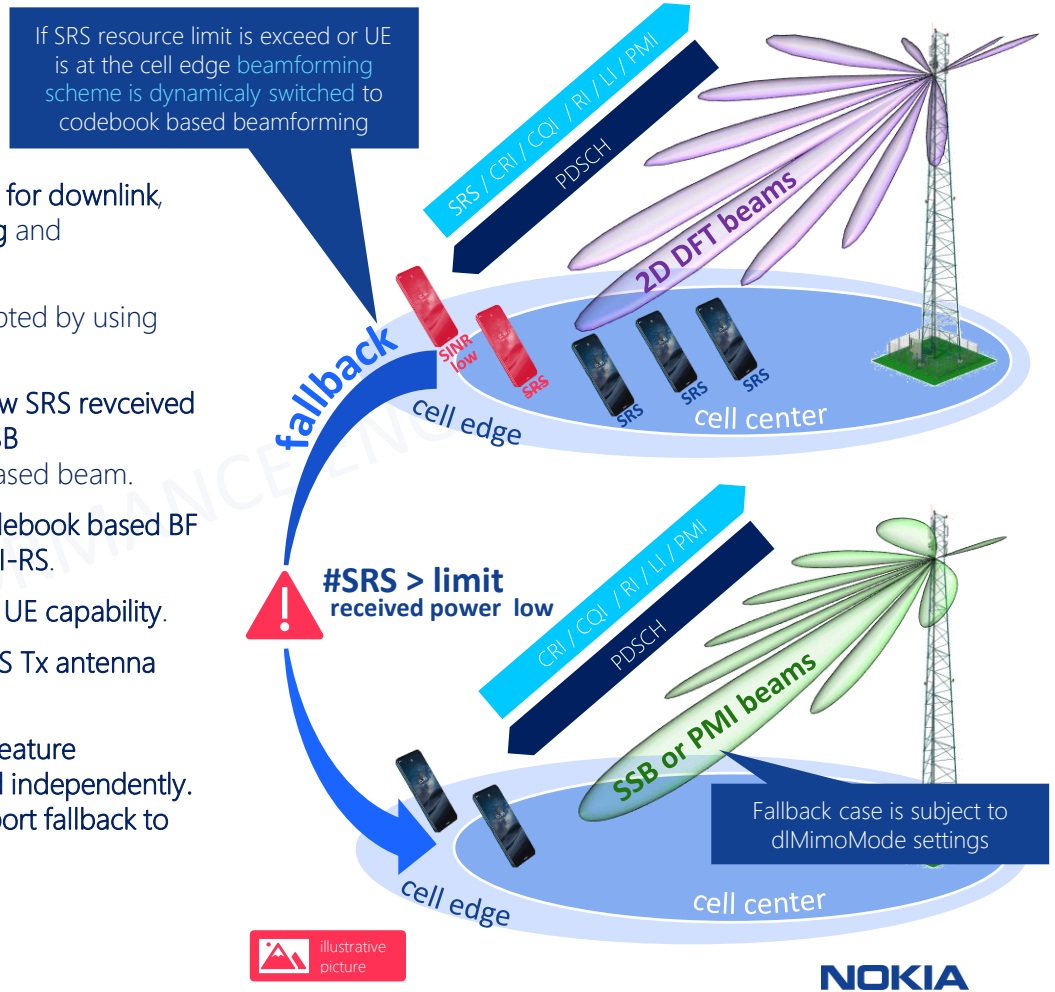
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Introduction

CB008247 feature overview

- This feature introduces the **dynamic beamforming schemes for downlink**, which means **dynamic switching of SRS based beamforming** and **codebook/SSB based** beamforming.
- For the **cell center** UE, **SRS based** beamforming will be adopted by using **covariance matrix calculation**.
- In case SRS covariance matrix cannot be available due to **low SRS revceived power**, **UE feedback based beam** can be applied such as **SSB** (5GC001856/5GC000605) or **8x4 codebook** (5GC002200) based beam.
- If **SRS resource is exhausted**, for other newly added UE, **codebook based BF** will be configured with **cell specific CSI-RS** or **UE specific CSI-RS**.
- UE will be configured with **SRS antenna switching** based on **UE capability**.
- **Codebook SRS** will be used in case **UE does not support SRS Tx antenna switching**.
- **CB008247 reuses** 5GC001086 SRS UL based beamforming **feature framework**, but CB008247 and 5GC001086 **can be activated independently.** However it is recommended to activate 5GC001086 **to support fallback to codebook SRS for UEs not supporting TAS SRS**.



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Introduction
CB008247 feature scope (1/2)

- The feature is applied to **TDD FR1 deployments** and to **32 TRX (including 32TRx split mode)** and **64 TRX RUs** (8 TRX RUs are out of feature scope).
- Supported **RU** types are **same as 5GC001086 SRS based beamforming in UL (WebNEI)**
  - nrCellType=**8DL4UL** required for **64TRX RUs**
  - nrCellType=**4DL4UL** and **8DL4UL** possible for **32TRX RUs**
- CB008247 applies to **eCPRI based RUs** supporting beam weight download via M-Plane (introduced by 5GC001866).
- The feature is **only based on NDM SRS***. For RUs not supporting NDM SRS, this feature shall be disabled, legacy beam management scheme is applied then.
- Both SA/NSA architecture options are supported
- UE Rel. 15 are supported, although Transmit Antenna Switching (TAS) SRS is UE optional feature
- The feature is **E2E feature**, it also includes impact on RUs.

*- Non Delay Managed SRS

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Introduction

CB010602 SRS based Beamforming extension

In 23R1 **CB010602 SRS based Beamforming extension** feature is introduced.

It enables to **balance the transmit power** between **weaker** and **stronger DL PDSCH beams** resulting in **improved DL throughput** performance of **rank 3 and rank 4 UEs**. Rank 1 and 2 UEs are not impacted.

Feature introduces parameters which allow to activate beam power balancing for cell-specific CSI-RS UEs and for UE-specific CSI-RS UEs

CB0010602 **replaces CNI-84878** which partially introduced beam power balancing in 22R4.

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Introduction
CB008247 feature scope (2/2)

- The ABIO/ASOE platforms are supported. ABIL is not in the scope.
- ABIN support is covered in later release by CB007134 ABIN NR legacy features support TDD eCPRI
- The feature does not support vRAN2.0 (a separate feature CB008608 is planned to cover vRAN for CB008247).
- CB008247 can be activated and deactivated by new introduced O&M NIDD parameter.
- There is no dependency between DL and UL beam management. CB008247 and 5GC001086 can be activated independently
- CB008247 supports an interworking with:
  - cDRX (gNB should consider the SRS report can be only achieved by gNB in DRX on mode).
  - CA (the SRS based DL beam will not be used for Scell).
  - EIRP and power saving (multiple BWPs).
- In addition, feature introduces CSI_RS multiplexing (activated by NRCELL. actCsirsMultiplexing parameter)

*- Non Delay Managed SRS

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Introduction

CB010388 Support of 5GC001086 and CB008247 using 4DL4UL celltype and 64TRX RUs

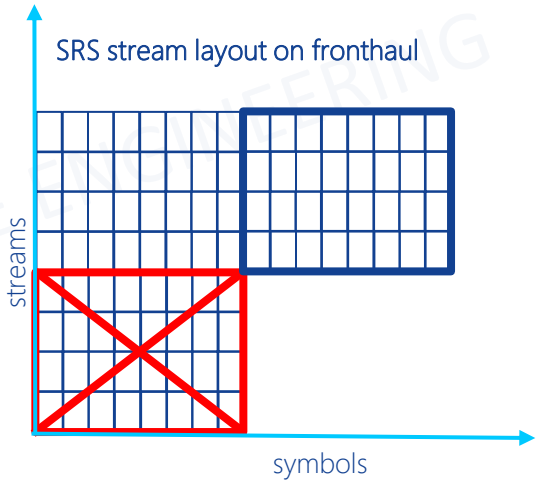
In 23R3 **CB010388 Support of 5GC001086 and CB008247 using 4DL4UL celltype and 64TRX RUs** feature is introduced.

This feature allows to run **SRS based beamforming features** like 5GC001086 and CB008247 for **nrCellType 4DL4UL** together with **64TRX RUs**.

Without this feature 64 TRX RUs required nrCellType=8DL4UL.

Feature works by changing the algorithm of **how the raw SRS signals are sent** from 64TRX RUs to baseband with the **reduced number of streams**.

No impact on SRS capacity or numer of RRC connected UEs is expected with this feature.



CB010388

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Introduction

CB010388 Support of 5GC001086 and CB008247 using 4DL4UL celltype and 64TRX RUs

For some cel sets SRS cannot be allocated with CB010388. These failre cases only happen when cells have **mixed nrCellType**, which is enabled by pooling features.

When it happens, the cell **can be setup** but NDM **SRS related features will be disabled** and **fault (5512) will be raised** on this cell.

The table lists all combinations of 3 cells (8DLnUL or 4DLnUL) with 64TRX or 32TRX radios. The cells with red rectangles are failure cases due to above limitation.

	Not supported configuration
	Supported (new configuration or timing is differerent with legacy)
	Supported (same as legacy)
	Supported (but cannot be tested in 23R2. It shall be covered by pooling features in 23R3)

Cells	nrCellType	TRX							
Cell#1	8	64	64	64	64	32	32	32	32
Cell#2	8	64	64	32	32	64	64	32	32
Cell#3	8	64	32	32	64	64	32	32	64
Cell#1	8	64	64	64	64	32	32	32	32
Cell#2	4	64	64	32	32	64	64	32	32
Cell#3	8	64	32	32	64	64	32	32	64
Cell#1	8	64	64	64	64	32	32	32	32
Cell#2	8	64	64	32	32	64	64	32	32
Cell#3	4	64	32	32	64	64	32	32	64
Cell#1	8	64	64	64	64	32	32	32	32
Cell#2	4	64	64	32	32	64	64	32	32
Cell#3	4	64	32	32	64	64	32	32	64
Cell#1	4	64	64	64	64	32	32	32	32
Cell#2	8	64	64	32	32	64	64	32	32
Cell#3	8	64	32	32	64	64	32	32	64
Cell#1	4	64	64	64	64	32	32	32	32
Cell#2	4	64	64	32	32	64	64	32	32
Cell#3	8	64	32	32	64	64	32	32	64
Cell#1	4	64	64	64	64	32	32	32	32
Cell#2	8	64	64	32	32	64	64	32	32
Cell#3	4	64	32	32	64	64	32	32	64
Cell#1	4	64	64	64	64	32	32	32	32
Cell#2	4	64	64	32	32	64	64	32	32
Cell#3	4	64	32	32	64	64	32	32	64

CB010388

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Introduction

CB010388 Support of 5GC001086 and CB008247 using 4DL4UL celltype and 64TRX RUs

Interaction with DL MU-MIMO (CB008305)

With **nrCellType=4DL4UL** the number of DL MIMO layers is 4.

This means that the maximum numer of UEs that can be multiplexed is 2, each with 2 DL MIMO layers.

To ensure adequate number of pairing candidates, parameter dlMuMimoRankAdaption needs to be set to **always** in order to force UE rank downgrade.

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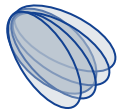
Introduction

Which beams do we have - so far?

5G18 5G19B 5G20A 22R2-5G 22R3-5G



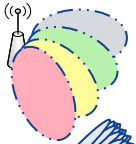
SSB beams: up to 8
In DL: all channels
In UL: all channels



UE compliance issues

refined beams: up to 32 (4 per SSB beam)
In DL: PDSCH
In UL: PUSCH

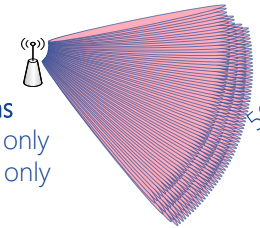
Introduced with 5GC002200



CSI-RS beams
In DL: cell-specific CSI-RS only

CNI-74278
Beamforming
Enhancements for 2200

PMI beams
In DL: PDSCH only



2D-DFT beams
In UL: PUSCH only
In DL: PDSCH only

5GC001086
UL SRS Beamforming

CB008247
DL SRS Beamforming

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Introduction
Supported Radio families

Please check most recent RU supporting CB008247 in Radio Unit Dashboard

Feature is supported by 32TRX and 64TRX RUs in FR1 TDD from RU families listed in the table

Product Family Name	Product Type	Product Names
Gen 4 mMIMO	Gen4 – 32 TRx mMIMO	AEQN
Chile	Gen4 – 64 TRx mMIMO	AEHB, AEHC
Master-eCPRI Upgrade	Made1 - 64TRx mMIMO - eCPRI	AEQB, AEQE, AEQP, AEQI, AEQJ, AEQK
Amber	Made1 - 32 TRx mMIMO	AEQV, AEQY, AEQZ, AEQQ, AEQC, AENB, AEHE
Megatron	Made2 - 64 TRx mMIMO	AQQL, AQQE, AQQP, AQQI, AQQK, AQQR
Galvatron	Made2 - 32TRx mMIMO	AQQY, AQQA, AQQQ, AQQZ, AQQC, AQHA

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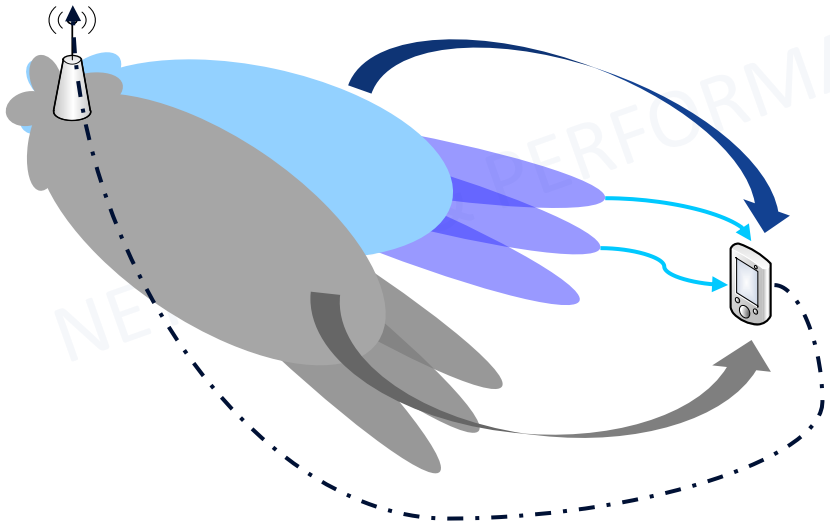
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Introduction

Legacy feature: 5GC000534 Digital Beamforming for eCPRI RUs

In 5GC000534 Digital Beamforming for CPRI Based RUs, beam management is based on:

- transmission of SSB beams and CSI-RS for beam refinement
- measurement and reporting (resource ID and RSRP) of best beams by UE



m e a s u r e

- SBRI-RSRP (SSB beams)
- CRI-RSRP (refined beams)

r e p o r t

- 2 best SBRI-RSRP
- 2 best CRI-RSRP

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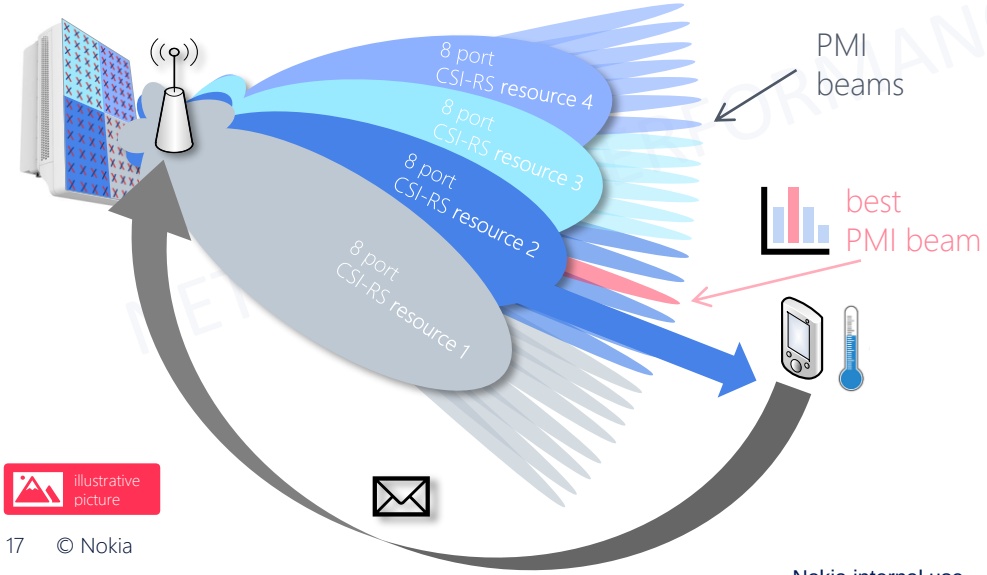
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Introduction

Legacy feature: 5GC002200 Support 4x8 port CSI-RS codebook for SU-MIMO

- Support **up to 4 beamformed CSI-RS resources**, where each CSI-RS **resource conveys 8 ports**.
- Firstly a **UE determines the best CSI-RS resource from plain power measurements** and hence chooses one of the 4 beams, e.g. it chooses beam with CSI-RS resource #2.
- This is a very robust choice since the CSI-RS are beamformed **and decision is only on power** without the need for error prone sophisticated math.
- Secondly a **UE choses a PMI beam using the 8-port codebook** e.g. in this case the red pencil beam within the resource 2 beam.
  The **UE reports CRI/CQI/RI/LI/PMI based on the selected best CSI-RS beam** using an 8-port codebook.
- **PBCH/PSS/SSS are transmitted on legacy SSB beams, feature is DL specific.**



UE measure
- CSI-RS (CSI-RS beams)

UE report
- CRI/RI/PMI

gNB select
- best PMI beam (up 2) for PDSCH

illustrative picture

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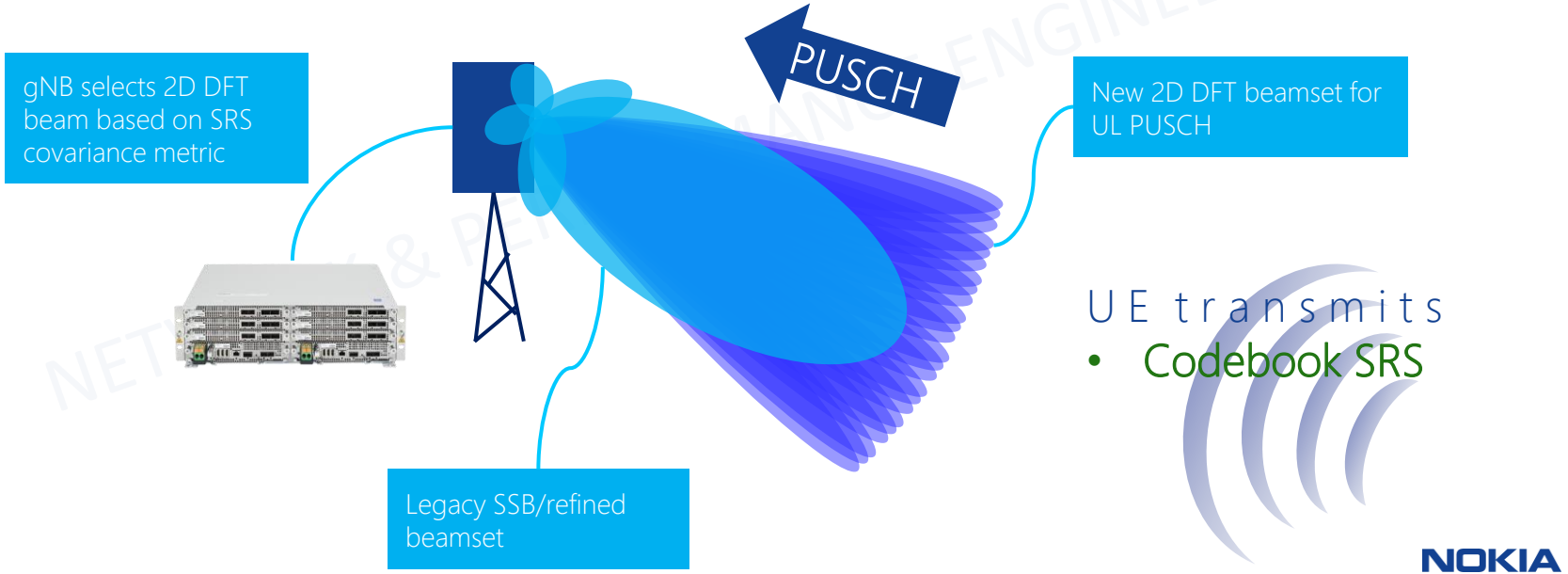
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Introduction

5GC001086 SRS based beamforming in UL

In 5GC001086 gNB processes UE SRS signal from **all TRXs** and selects a beam from 2D DFT beamset based on SRS covariance.



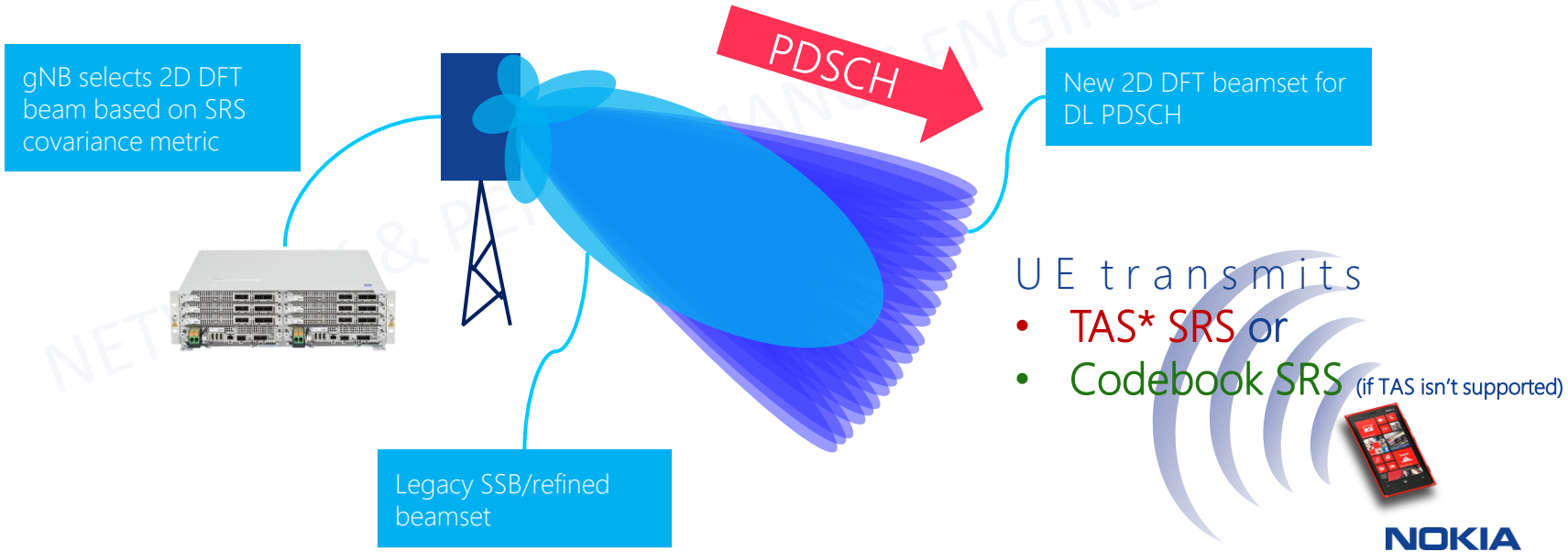
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Introduction

THIS FEATURE: CB008247 SRS based beamforming in DL

In 5GC008247 gNB processes UE SRS signal from **all TRXs** and selects a beam from 2D DFT beamset based on SRS covariance.



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* - TAS: Transmit Antenna Switching

Introduction
Before & after (CB008247)

Before

- DL beamforming methods in FR1-TDD include
  - Legacy SSB (optionally with beam refinement)
  - PMI beams
- In UL, SRS based beamforming available in 22R2

After

- DL SRS based beamforming introduced in FR1-TDD
- Improvement in average DL cell and user throughput

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Introduction
Before & after (CB010602)

Before

- Power Balancing between weaker and stronger beams not present (22R3) or limited (in 22R4)
- Performance loss in for rank 3 and rank 4 UEs

After

- Power balancing available for all rank 3 and rank 4 UEs
- Improved feature DL throughput performance

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Technical Details

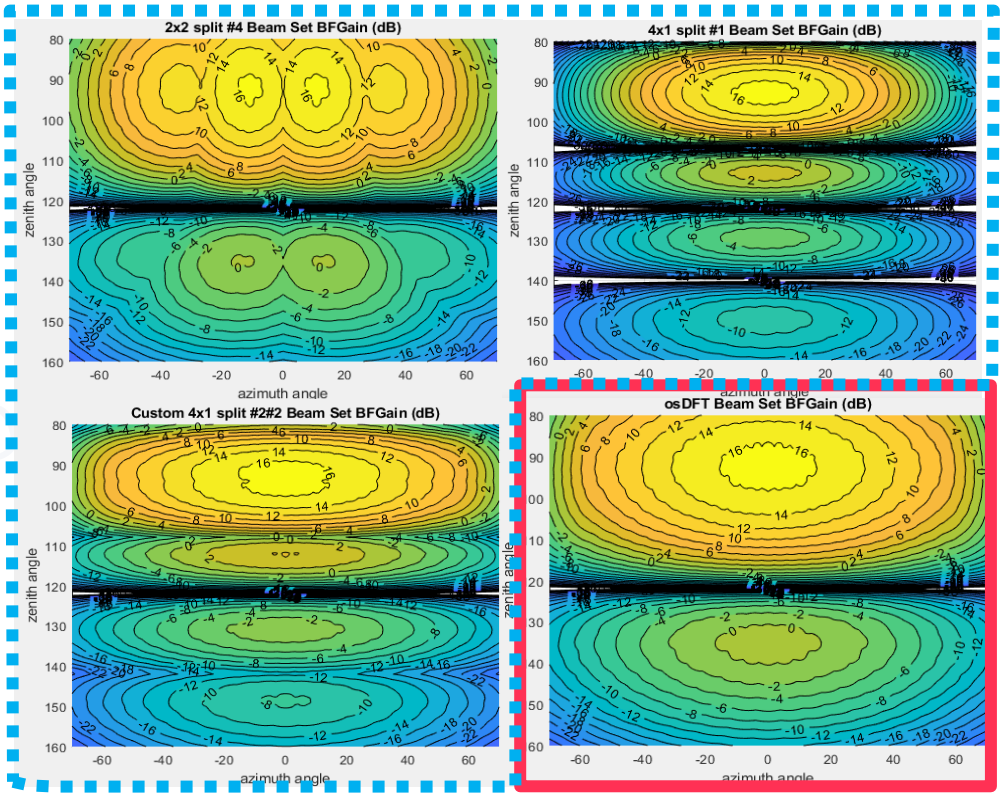


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Technical Details

CB008247– where the gain comes from?

Example, 32TRX RU



CB008247 is using SRS to derive **additional information** on the channel, in order to **enhance DL transmission**.

It is based on 2D DFT beamset introduced by 5GC001086. Best beams are chosen from that beamset using SRS information.

Compared to PMI beamset introduced by 5GC002200, new beamset provides higher beam resolution and flexibility:

- More beams – higher resolution;
- Additional SRS information – better beam adjustment compared to 5GC002200

PMI beamset (5GC002200)

2D DFT beamset (CB008247)

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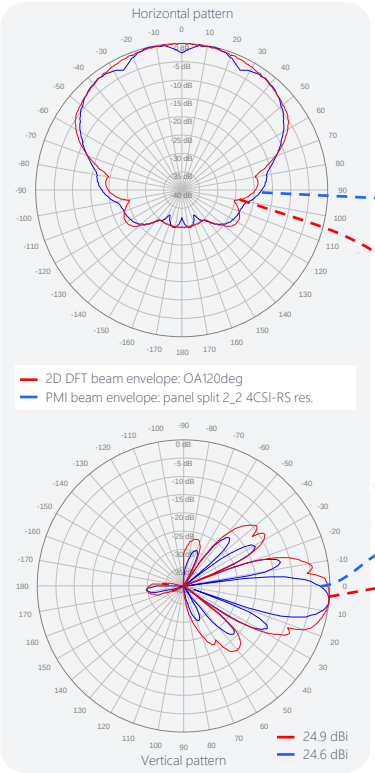
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illustrative picture

Coverage Aspects of 2D DFT Beams vs PMI Beams (2_2 #4 CSI-RS Resources)

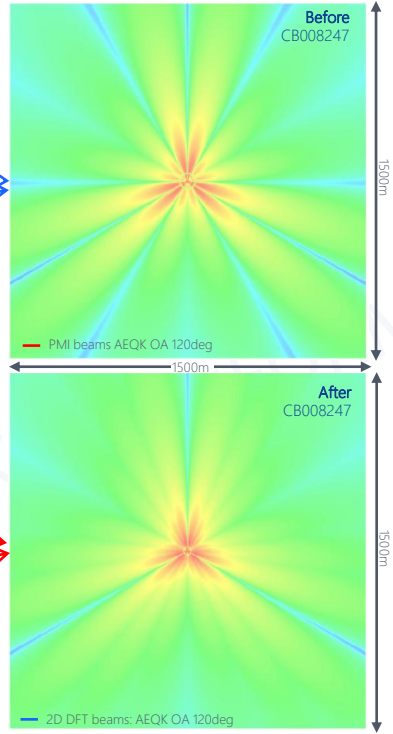
Example composite signal coverage footprint relative comparison, RU AEQK

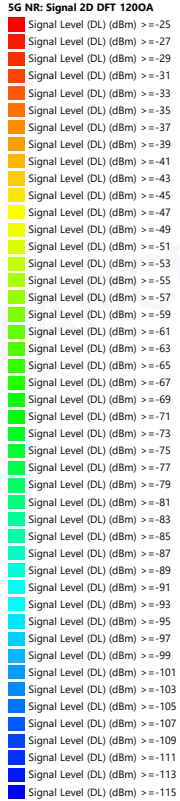
2D DFT and PMI beam envelope patterns



RU AEQK 64TRX, 192AE, Max power: 53dBm

Composit signal coverage



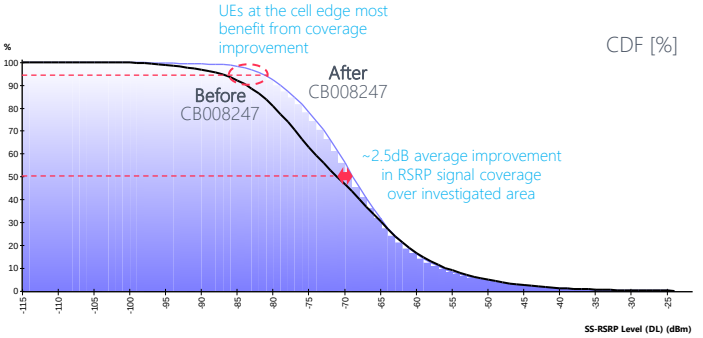


3-sector site, azimuths: 0°,120°,240°, antenna height: 25m AGL, propagation model: SPM DU tuned at 3.5GHz, mapping data 5m: flat DTM 0m, homogenous land usage

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Introduction of CB00824700 may help to improve coverage due to much more dense design of 2D DFT beams, conveying data streams.

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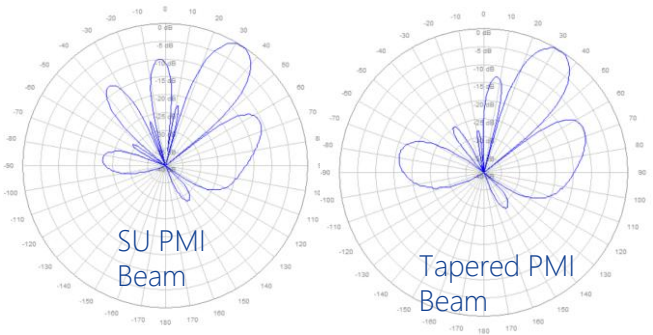
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Technical Details

CB008247 Benefits over 5GC002200

- Compared to legacy PMI beams introduced by 5GC002200 there **is better sidelobe expression** of the new 2D DFT beamset;
- **More beams in 2D DFT beamset** used by CB008247 improve beam resolution compared to PMI beams (UE can be reached with higher precision/higher gain)
- 8247 allows **more flexible use of beams** – angular separation of the two beam can be **arbitrarily large** while fixed for PMI beams
- **Better interference estimation** with SRS is also an improvement required by CB008305 MU-MIMO feature, which can be further improved by **beam tapering**



5GC002200 2x2 PMI beam



CB008247 DFT beam

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Nokia internal use

Technical Details

2D DFT beamset

Instead of using predefined beamset weights as introduced by 5GC00533/534 or 5GC002200, weights of the 2D DFT GoB are calculated internally by gNB based on the deployed RU antenna geometry

- number of TRX columns/ rows
- vertical/ horizontal distance of the radiating elements
- operating frequency wavelength

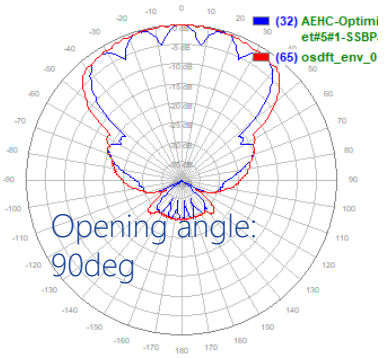
...and basic SSB beamset parameters:

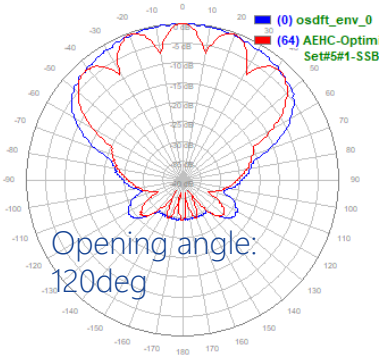
- tiltOffset
- leftEdgeAngle/ rightEdgeAngle
- upperEdgeAngle/ lowerEdgeAngle

The algorithm adjusts the number of 2D DFT beams in vertical and horizontal planes to match the coverage of the SSB beamset.

Note: Different (higher) numer of 2D DFT beams in 22R3 compared to 22R2. 5GC001086 will also use new beamsets. Possible impact on UL performance in 22R3 even if CB008247 not activated.

RU type	TRX colu mns	TRX rows	Horiz. opening angle [deg]	2D DFT columns	2D DFT rows	Number of 2D DFT beams
64TRX AEHC	8	4	90	23	14	322
			120	27	14	378
64TRX AEQK	8	4	90	23	14	322
			120	29	14	406
32TRX AEQZ	8	2	90	21	8	168
			120	27	8	216
32TRX AEQN	8	2	90	23	7	161
			120	29	7	203





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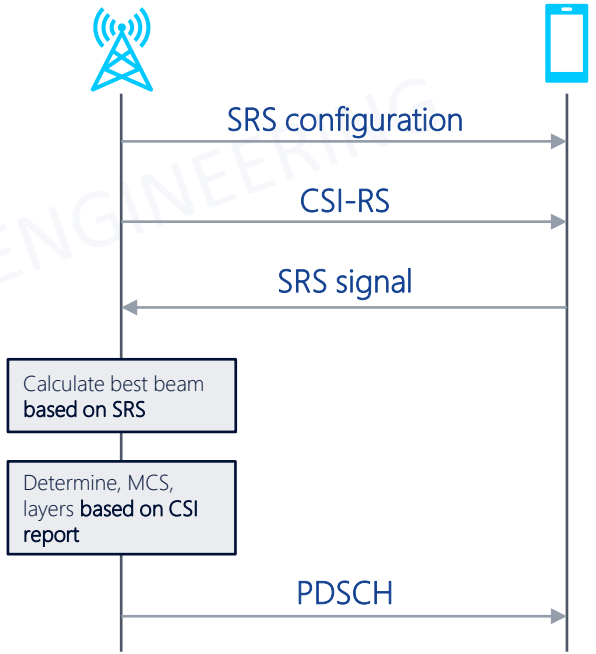
Nokia internal use

Technical Details

CB008247 processing chain overview

In high level, the processing chain is shown below:

- **gNB configures SRS resource to UE** during UE setup. In this feature only periodic SRS is configured for beam management.
- **Cell specific CSI-RS** or **UE specific CSI-RS is configured** to UE during UE setup.
- **UE transmits SRS based** on gNB configuration
- **gNB calculates the best beam(s) for PDSCH** data transmission.
- **gNB determines the MCS** and data layers based on CSI report.
- **PDSCH is transmitted** with the best beam(s).



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Technical Details

SRS configuration

- Same as 5GC001086, the SRS for DL beam management is only configured in special slot
- The SRS symbol position is determined according to frame structure and special slot configuration.
- Depending on frame structure following order of special slot placement is given below:
  - 5GC001208: #3,#8,#13,#18
  - 5GC001070: #3,#7,#13,#17
  - 5GC001116_0ms: #3,#13
  - 5GC001116_3ms: #7,#17

Periodic SRS is only allocated in special slots

Subframe	0		1		2		3		4		5		6		7		8		9	
Slot	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
1208	D	D	D	S	U	D	D	D	S	U	D	D	D	S	U	D	D	D	S	U
1070	D	D	D	S	U	D	D	S	U	U	D	D	D	S	U	D	D	S	U	U
1116_0ms	D	D	D	S	U	U	D	D	D	D	D	D	D	S	U	U	D	D	D	D
1116_3ms	D	D	D	D	D	D	D	S	U	U	D	D	D	D	D	D	D	S	U	U

Special slot placement in different subframe formats
D Downlink U Uplink S Special slot

SRS periodicity: default value: 40 slots (srsForBeamMgmt.srsUePeriodicity)
SRS cell BW: channel BW dependent, set by (srsForBeamMgmt.cSrs)
SRS subband size: channel BW dependent, set by (srsForBeamMgmt.bSrs)

Note: depending on the configured TDD FR1 frame structure NRCELL.frameStructureType (semiStatic or tdLte) and configured NRCELL.guardPeriodLength, this SRS symbol position in special slots may be any position among symbol index {10,11,12,13}.

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Technical Details

SRS configuration basics: Frequency Allocation (1)

SRS resource is configured to UE and consists of:

- Number of SRS antenna ports {1, 2, 4}
- Cyclic shift and comb
- **Frequency allocation**
  - Bandwidth
  - Offset relative to start of BWP
  - Frequency hopping activation
- Time location
  - Starting OFDM symbol
  - Number of OFDM symbols
  - Repetition factor
  - CAZAC sequence initialization
- Resource type {aperiodic, semiPersistent, periodic}

- Not configured *directly*
- NRCell.cSRS and NRCEll. bSRS NIDD parameters located under structure NRCELL.srsForBeamMgmt are used **to configure SRS bandwidth and frequency domain position**.
- Both parameters determine which **portion of BWP is sounded by SRS resource**.
- NRCell.cSRS and NRCEll.bSRS are used **to identify a row and column in the lookup table** (*3GPP 38.211 Table 6.4.1.4.3-1*).
- Each configuration of **NRCell.cSRS and NRCEll.bSRS** is related to given **set of ($m_{SRS,b}$,N_b)** parameters from the table, where b index: {0,1,2,3} .
- For given NRCell.cSRS settings **$m_{SRS,0}$ determines maximum bandwidth within BWP** that can be sounded by configured SRS resource.

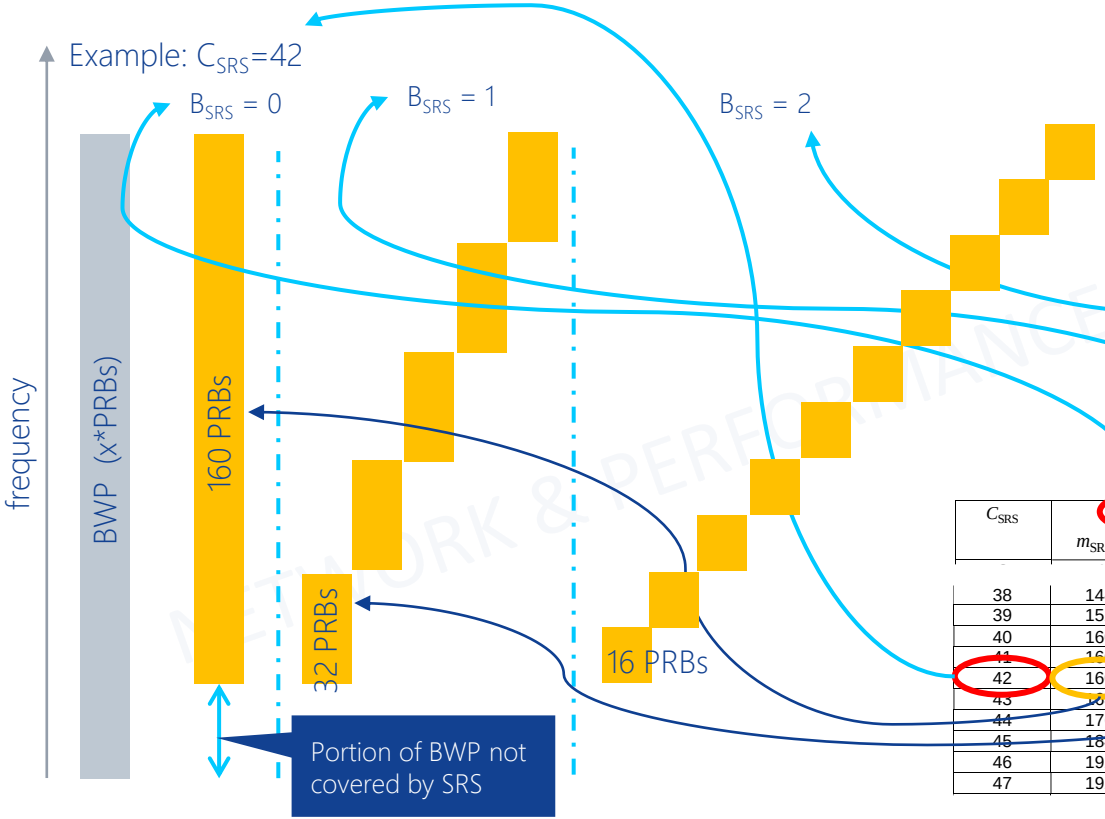
C_{SRS}	$B_{SRS}=0$		$B_{SRS}=1$		$B_{SRS}=2$		$B_{SRS}=3$	
	$m_{SRS,0}$	N_0	$m_{SRS,1}$	N_1	$m_{SRS,2}$	N_2	$m_{SRS,3}$	N_3
0	4	1	4	1	4	1	4	1
1	8	1	4	2	4	1	4	1
2	12	1	4	3	4	1	4	1
3	16	1	4	4	4	1	4	1
4	16	1	8	2	4	2	4	1
5	20	1	4	5	4	1	4	1
6	24	1	4	6	4	1	4	1
7	24	1	12	2	4	3	4	1
8	28	1	4	7	4	1	4	1
9	32	1	16	2	8	2	4	2
10	36	1	12	3	4	3	4	1
11	40	1	20	2	4	5	4	1

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Technical Details

SRS configuration basics: Frequency Allocation (3)



C_{SRS}
- row index in 3GPP table
- provides wideband SRS allocation span
- defines subband split options

B_{SRS}
- column index in 3GPP table
- provides actual SRS bandwidth

C_{SRS}	$B_{SRS}=0$		$B_{SRS}=1$		$B_{SRS}=2$		$B_{SRS}=3$	
	$m_{SRS,0}$	N_0	$m_{SRS,1}$	N_1	$m_{SRS,2}$	N_2	$m_{SRS,3}$	N_3
38	144	1	16	9	8	2	4	2
39	152	1	76	2	4	19	4	1
40	160	1	80	2	40	2	4	10
41	160	1	80	2	20	4	4	5
42	160	1	32	5	16	2	4	4
43	168	1	84	2	28	3	4	7
44	176	1	88	2	44	2	4	11
45	184	1	92	2	4	23	4	1
46	192	1	96	2	48	2	4	12
47	192	1	96	2	24	4	4	6

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Technical Details

Nokia product specific SRS bandwidth configurations

New recommendation

1st candidate								2nd candidate					
Channel Bandwidth (chBw)	PRB amount	SRS Bandwidth (cSRS)	SRS Subband (bSRS)	SRS frequency range	Subband size	SRS Capacity per slot	SRS vs BW	SRS Bandwidth (cSRS)	SRS Subband (bSRS)	SRS frequency range	Subband size	SRS Capacity per slot	SRS vs BW
20	51	12	2	48	8	48	94.12%						
30	78	19	2	72	12	48	92.31%	20	2	76	4	152	97.44%
40	106	23	2	96	16	48	90.57%	25	2	104	4	208	98.11%
50	133	32	2	128	8	128	96.24%						
60	162	38	2	144	8	144	88.89%	42	2	160	16	80	98.77%
70	189	42	2	160	16	80	84.66%	45	2	184	4	368	97.35%
80	217	49	2	192	8	192	88.48%						
90	245	55	3	240	8	240	97.96%						
100	273	63	2	272	8	272	99.63%						

- The defined values represent balance between the SRS capacity and performance.
- If no specific requirement e.g. single UE peak throughput test, then 1st candidate should be used as default settings.

NOTE:

- SRS subband size **8-16 PRB** is **the first recommendation**
- **4 PRB works fine in simulation** and is available for field trials.
- **Smaller SRS** subband size provides **higher SRS capacity** and **wider coverage**. 4PRB can be used if it works fine in field after testing

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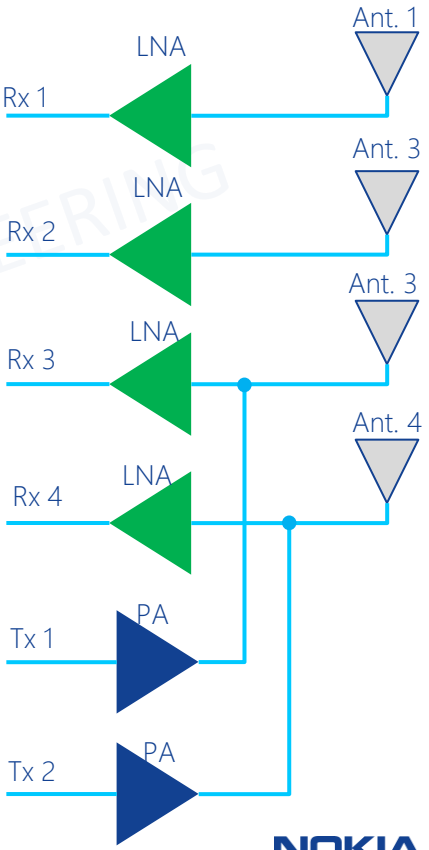
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Technical Details
Codebook SRS Principle

- For practical reasons the number of **UE transmit antennas** is **smaller in UL** than in DL:
  - UEs are power limited
  - Implementation issues - RF placement, specific absorption rate (SAR)
  - device cost
- For **UL MIMO purposes** 1 or 2 SRS resources can be **configured to the UE**
- When 2 resources are configured SRS will be used to derive the precoding for UL transmission according to 3GPP 38.211 tables 6.3.1.5-1.. 6.3.1.5-7
- This type of SRS is referred to as **codebook SRS**

```
SRS-ResourceSet ::= SEQUENCE {
  srs-ResourceSetId          SRS-ResourceSetId,
  srs-ResourceIdList         SEQUENCE (SIZE(1..maxNrofSRS-ResourcesPerSet))
                                 OF SRS-ResourceId OPTIONAL, -- Cond Setup
  [...]

  usage                        ENUMERATED {beamManagement, codebook, nonCodebook, antennaSwitching},
  alpha                    Alpha OPTIONAL, -- Need S
  p0                       INTEGER (-202..24) OPTIONAL, -- Cond Setup
  pathlossReferenceRS CHOICE {
     ssb-Index             SSB-Index,
     csi-RS-Index          NZP-CSI-RS-ResourceId
  } OPTIONAL, -- Need M
  srs-PowerControlAdjustmentStates        ENUMERATED { sameAsFci2, separateClosedLoop} OPTIONAL,
  ...
}
```



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Technical Details

TAS principle

In Transmit Antenna Switching, UE is transmitting according to its switching capability indicated by *supportedSRS-TxPortSwitch* parameter:

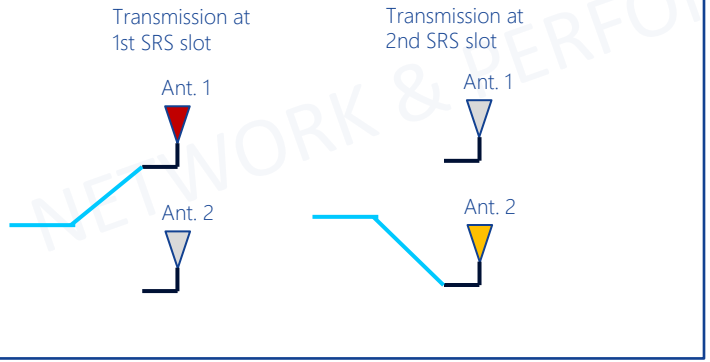
supportedSRS-TxPortSwitch = mTnR

where:
m - number of antennas transmitting SRS in the same symbol
n - total numer of SRS transmitting antennas

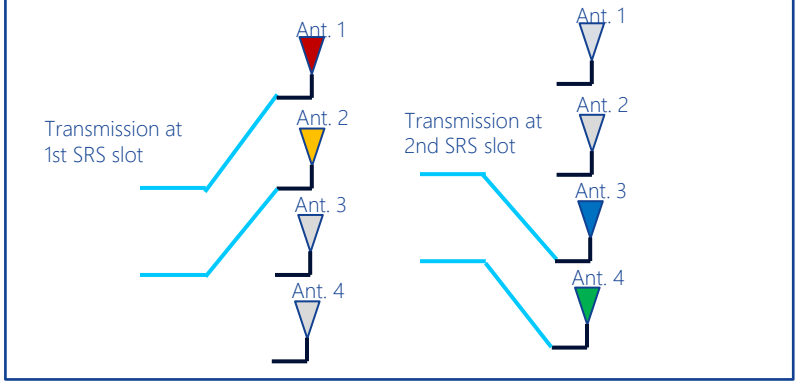
TAS SRS benefit over codebook SRS:

- TAS SRS allows **sounding of all DL RX** antennas used by UE which allows better channel estimation for DL SU-MIMO.
- Codebook SRS allows sounding **only UL TX** antennas used by UE.
- **TAS SRS require more SRS resources** compared to codeboook SRS, which may result in lower numer of UEs that can be configured with DL SRS beamforming.

Example: supportedSRS-TxPortSwitch=**1T2R**



Example: supportedSRS-TxPortSwitch=**2T4R**



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Technical Details

SRS resource management and configuration: SRS **T**ransmit **A**ntenna **S**witching (TAS)

- In 5GC001086, the codebook usage periodic SRS is supported.
- With CB008247 feature, the antenna switching usage periodic SRS (TAS SRS) will be supported for DL beam management.
- TAS SRS is UE optional capability and is indicated by *supportedSRS-TxPortSwitch* information field:

antenna switching capability	antenna switching SRS configuration		codebook SRS configuration
supportedSRS-TxPortSwitch	resource number	ports per resource	ports per resource
t1r2	2	1	1
t1r4	4	1	1
t2r4	2	2	2
t1r1	1	1	1
t2r2	1	2	2
t4r4	1	4	4
t1r4-t2r4	4 or 2	1 or 2 (see notes)	1 or 2

if maxNumberSRS-Ports-PerResource= 1, consider it as T1R4, otherwise consider it as T2R4

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Technical Details

SRS resource allocation principle

NIDD NRCELL.srsForBeamMgmt.**srsTypeAllocForDlBm** parameter allows to configure type of SRS used for SRS based DL beamforming.
3 options are available to configure: (0) **antenna switching** / (1) **codebook** / (2) **antenna switching or codebook** (default)

- The TAS SRS is only **configured in special slot** and **only 1 symbol** is used for NDM SRS due to L1 processing capability limitation.
- The TAS SRS **bandwidth and periodicity** are controlled by **NIDD parameters**.
- **If there are free SRS resources in the cell**, the SRS is configured to UE based on UE capability during UE accessing network after UE capability is available:
  - **If** UE supports antenna switching, **configure 2 SRS resource sets**. One set has 1 resource with **codebook for UL beam management and UL SU-MIMO**. Another set has **multiple resources with "antenna switching usage" for DL beam management**. The SRS resource number and ports are based on UE capability.
  - **Else**, configure **only one SRS resource set** with "codebook" for UL beam management, UL SU-MIMO and DL beam management
- If there are **no free periodic SRS resources**, only **aperiodic SRS with codebook** is assigned to UE which is used for UL SU-MIMO*. SRS based DL beam management is **not applied to this UE.**

TAS SRS is used for **DL beam management only**

Codebook SRS is used for **UL beam management/UL SU- MIMO** (as in 5GC001086)

If UE does not support **TAS SRS**, **Codebook SRS** will be used for **DL beam management** as well

If there are **no SRS resources** to allocate to UE, SRS based beamforming is **not used.** **Aperiodic SRS** are **still configured** for **UL SU-MIMO***



It is recommended **to activate 5GC001086** to support fallback to codebook SRS for UEs not supporting TAS SRS

* - more details on UL SU-MIMO in webNEI: 5GC000532

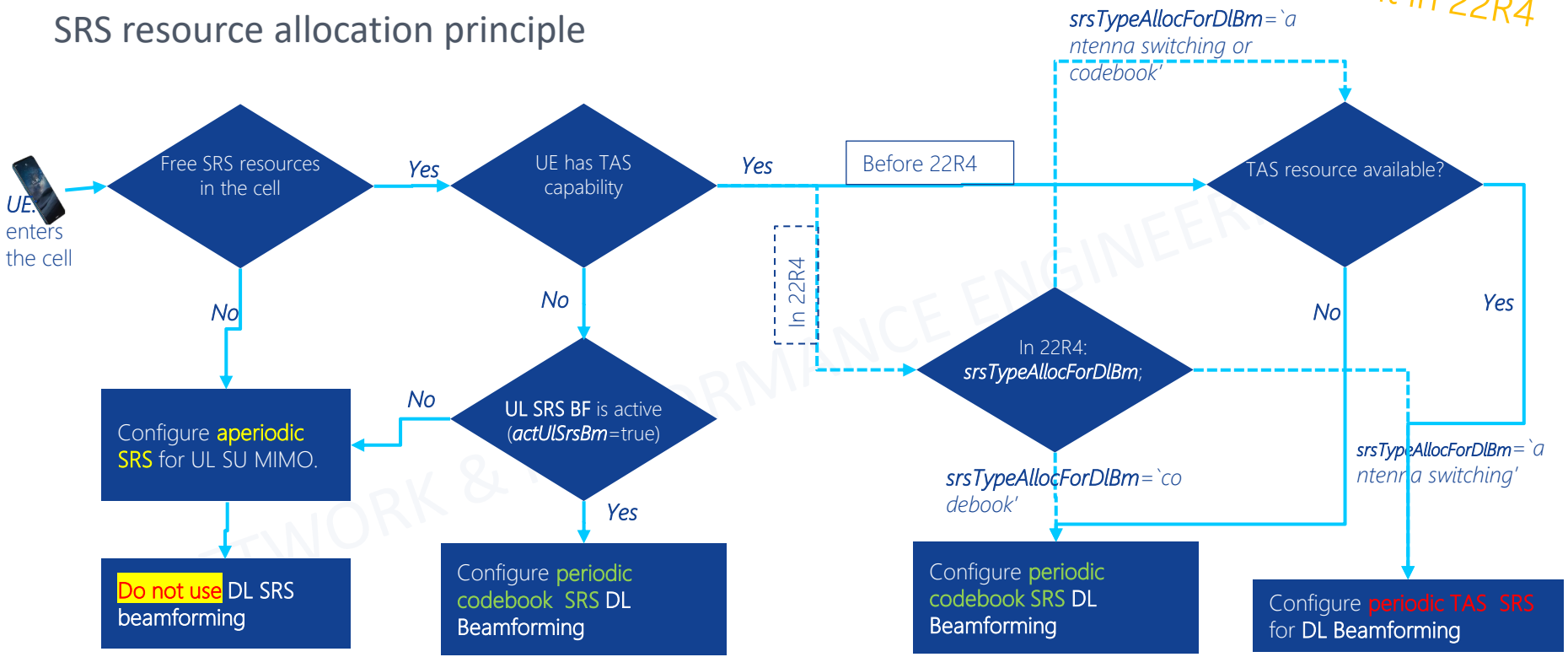
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Technical Details

SRS resource allocation principle

New improvement in 22R4



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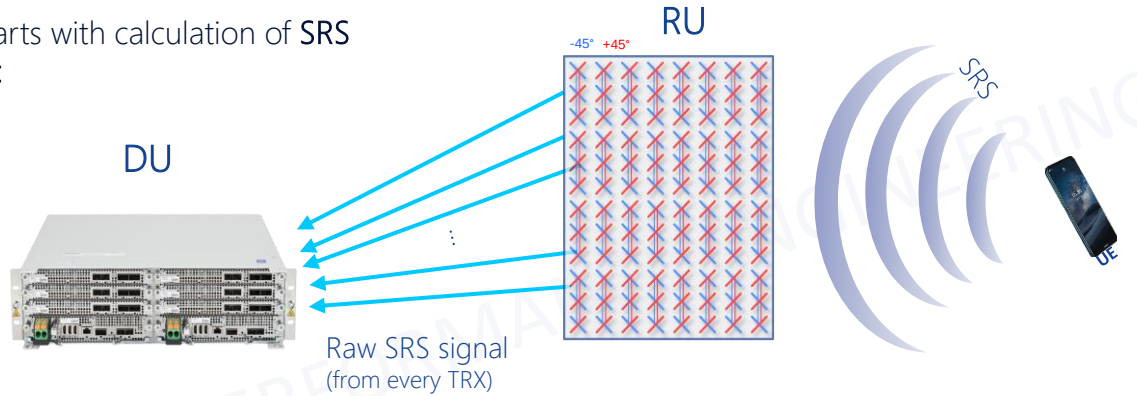
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Technical Details

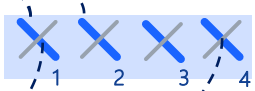
Principle of SRS-based beam selection (1)

illustrative picture

Best beam determination starts with calculation of **SRS covariance matrix in the DU:**



$$R_{SRS} = \begin{bmatrix} h_1h_1^* & h_2h_1^* & h_3h_1^* & h_4h_1^* \\ h_1h_2^* & h_2h_2^* & h_3h_2^* & h_4h_2^* \\ h_1h_3^* & h_2h_3^* & h_3h_3^* & h_4h_3^* \\ h_1h_4^* & h_2h_4^* & h_3h_4^* & h_4h_4^* \end{bmatrix}$$



Example:
4 antennas, 1 polarization

h_n: SRS received on n-th receiving antenna

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Technical Details

Principle of SRS-based beam selection (2)

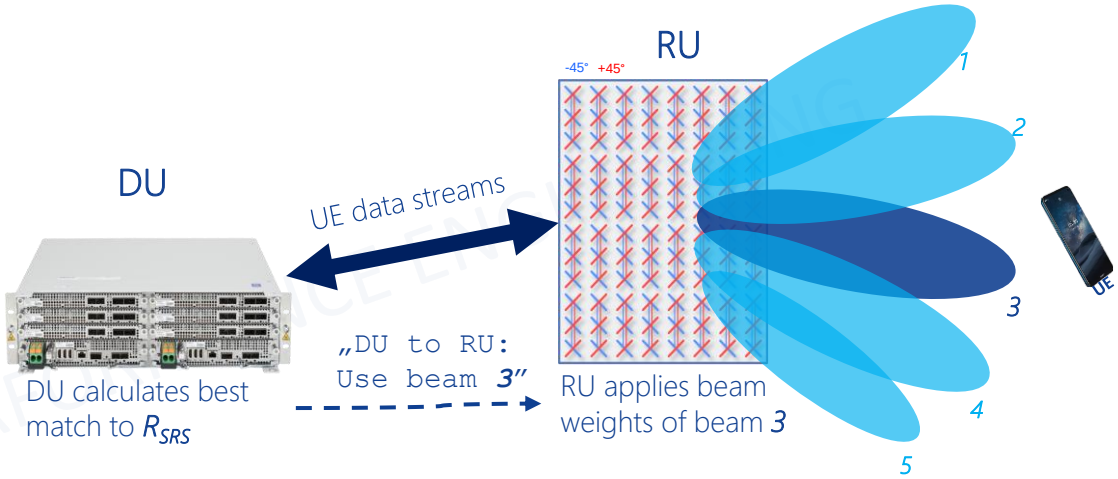
illustrative picture

Check which **beamforming vector** *w* provides **highest result of**:

$$w_k^H \cdot R_{SRS} \cdot w_k$$

SRS **covariance matrix**

Beamforming vector of beam *k*



From the 2D DFT beamset **best beam** is selected on the basis of „**best match**" to the received SRS signals.

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Technical Details

Selecting the **Best** and **Second Best Beams**

The **best and 2nd best beams** need to be determined per UE



In UL:

For **4RX reception with beamforming**, **2 beams** need to be determined per UE.

Algorithm selects the 2nd best beam on the similar basis as the best beam, however the **correlation** between the **best** and **2nd best** beam is also checked. 2nd best beams **too closely correlated** to best beam **are rejected.**

In DL:

2 beams are needed for **Rank 3** and **Rank 4** transmissions.

Due to different requirements in DL, the algorithm searches for the optimum beams **jointly**.

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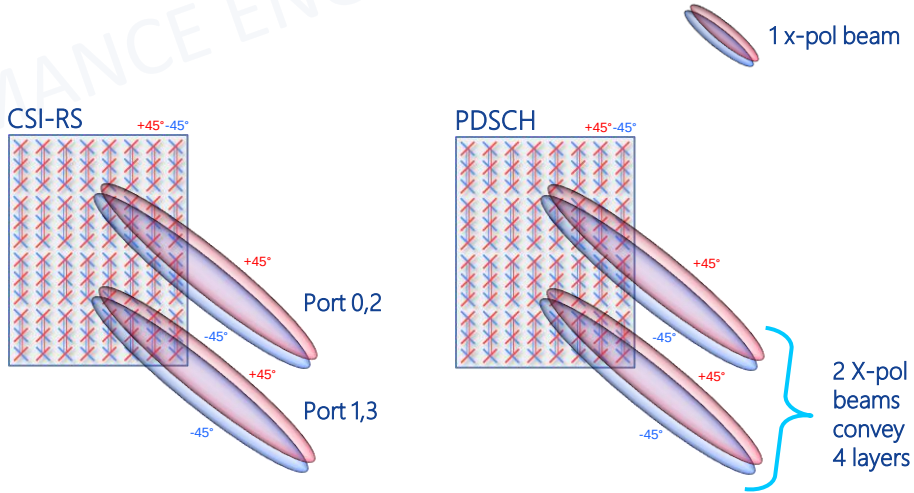
Technical Details

DL beam management (1/3)

- Depending on CSI-RS configuration for channel acquisition, there are two types of UEs: UEs configured with **UE-specific CSI-RS** for channel acquisition, and UEs configured with **cell-specific CSI-RS** for channel acquisition.
- **Cell specific CSI-RS** is only configured to UEs in case of **8x4 CL-MIMO**. In **other cases, UE specific CSI-RS** is configured to UEs.
- If **aperiodic SRS** is configured to UEs, then the **legacy scheme for beam management** is applied.
- If **periodic SRS** is configured to UEs, then two cases are supported:

Case 1: Periodic SRS + UE specific CSI-RS

- CSI-RS:
  - 4 ports mapped to **2 beams** with **2 polarizations**
  - 1 or 2 best beams got from **pSRS measurement**
- PDSCH:
  - Layer 1, 2: use **1 beam with 2 polarizations**
  - Layer 3, 4: use **2 beams with 2 polarizations**
- PDCCH:
  - Only the **best beam** is applied.



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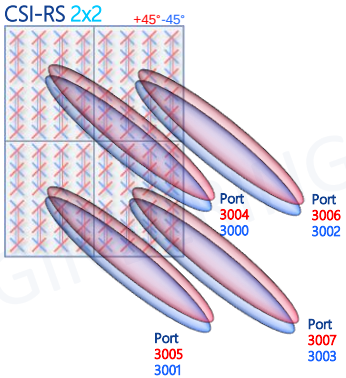
Technical Details

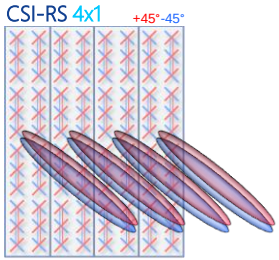
DL beam management (2/3)

… continued

Case 2: Periodic SRS + cell specific CSI-RS

- In 5GC002200, 2 different cell specific CSI-RS configurations are supported: 2x2 subpanel split and 4x1 subpanel split as shown below. 8 ports are mapped to the 4 subpanels with 2 polarizations.
- In CB008247, the same cell specific CSI-RS configurations are applied.
- For PDSCH, to support up to 4 layers, up to 2 best beams are got from periodic SRS measurement. Only i_2 in PMI reported by UE is used to generate the phase offset between two polarizations
- For PDCCH, the best beam is applied.





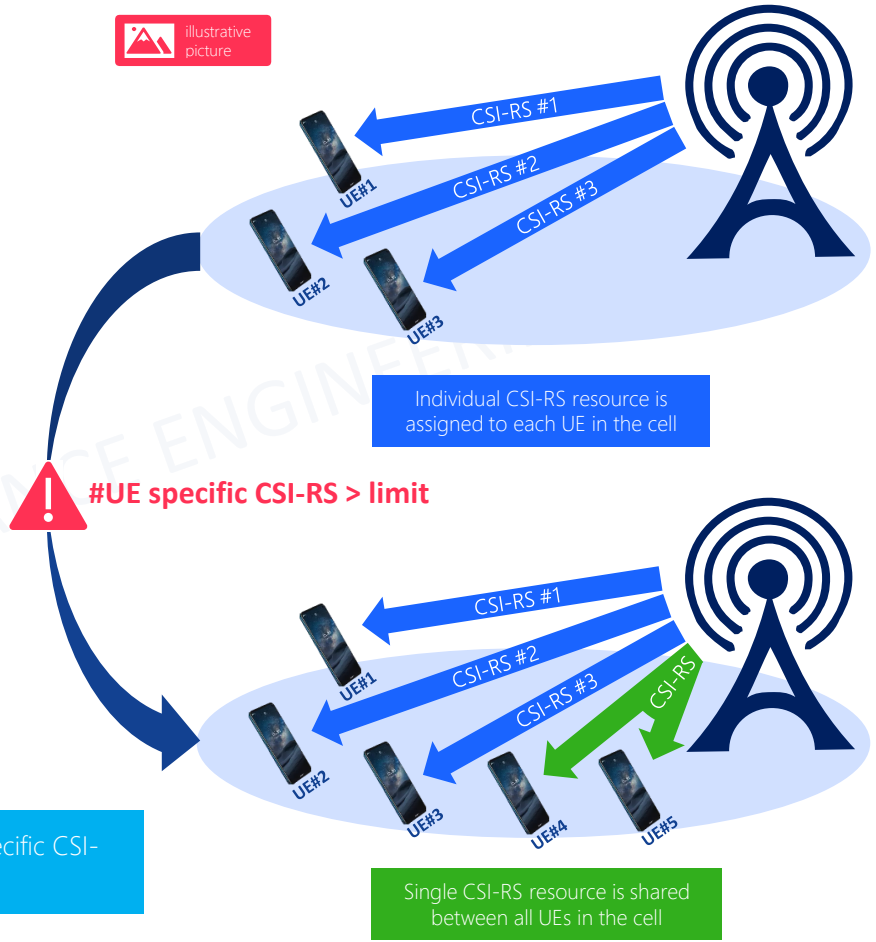
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Technical Details
DL beam management (3/3)

... continued

Selection of CSI-RS resource

- If dlMimoMode is equal to '8x4 Closed Loop Spatial Multiplexing' (5GC002200 feature is enabled) the activation flag NRCELL. actUeCsirsCoexistCellCsirs enables coexistance of UE specific CSI-RS with cel specific CSI-RS.
- If the activation flag is set to TRUE then the gNB will firstly allocate UE specific CSI-RS resources.
- If UE is allocated with periodic SRS, then UE is configured with UE specific CSI-RS. If UE cannot be configured with UE specific CSI-RS, then cell specific CSI-RS configurations is chosen.



With CB008247 performance with UE-specific CSI-RS generally is better than with cell-specific CSI-RS. See next slides for exception examples.

Technical Details

High BLER issue observed with CB008247 with actUeCsirsCoexistCellCsirs enabled

- High PDSCH BLER (> 40%) on SSB slot (slot0) compared to other slots with PDSCH was observed after activation of DL SRS BF (CB008247) with actUeCsirsCoexistCellCsirs = true (UE specific CSI-RS in use)
- In order to overcome this issue CNI-92005 *No PDSCH scheduling in last symbol of SSB slot to mitigate high BLER* has been introduced.
  - Suspection: interference from UE specific CSI-RS of neighbor cells
  - Mitigation proposal: Exclude the last PDSCH symbol of SSB slot from scheduling
  - Knife has been created and verified from commercial sites
  - CNI introduces knife content to trunk with feature flag to allow enabling only for requesting customers

Analysis done:

Typical transmission on SSB slot: NRB=225, MCS=22, Num layer=3

- With W/A function off, TBS on SSB slot : 524640 bits
- With function on, TBS on SSB slot : 450984 bits

Data volume on SSB slots per one second:

- W/A off : 524640 bits x (1000ms/20ms) x 0.6 (BLER 40%) = 15739200 bits
- W/A on : 450984 bits x (1000ms/20ms) x 0.8 (BLER 20%) = 18039360 bits (14% ↑)

NETWORK & PERFORMANCE ENGINEERING

Technical Details

Initial issue description

- High PDSCH BLER (> 40%) on SSB slot (slot0) compared to other slots with PDSCH was observed after activation of DL SRS BF (CB008247) with actUeCsirsCoexistCellCsirs = true (UE specific CSI-RS use).
- DL BLER is not severely high at the SSB0 slot where UE specific CSI-RS is allocated for the test UE -> Evidence that high DL BLER of other SSB0 slots caused by interference from UE specific CSI-RS on SSB slots of neighbor

[actUeCsirsCoexistCellCsirs = false]

DL BLER	slot	Slot																			
	0		1		2		3		4		5		6		7		8		9		
sfn_160ms	D	D	D	S	U	D	D	D	S	U	D	D	D	S	U	D	D	D	S	U	
0	12.95	25.00	29.59	5.80	0.00	28.55	6.90	11.19	5.87	0.00	23.74	15.67	6.79	6.48	0.00	6.56	4.84	4.94	4.90	0.00	
1	8.93	5.10	0.00	0.00	0.00	10.15	5.64	6.13	4.74	0.00	8.07	5.81	5.72	4.80	0.00	7.13	6.01	5.59	4.86	0.00	
2	9.34	4.55	6.12	5.31	0.00	8.49	5.70	5.49	4.87	0.00	9.78	6.85	8.62	45.21	0.00	10.92	10.99	7.06	15.45	0.00	
3	10.42	7.52	6.99	11.92	0.00	9.38	6.65	7.13	6.24	0.00	10.36	6.85	5.24	5.14	0.00	8.39	6.24	18.37	4.86	0.00	
4	15.75	5.53	6.05	4.74	0.00	25.01	6.08	6.39	5.44	0.00	8.97	6.03	5.71	5.85	0.00	8.27	6.37	5.47	5.53	0.00	
5	8.65	6.27	0.00	0.00	0.00	10.98	6.72	7.21	4.57	0.00	9.22	6.86	6.55	6.57	0.00	27.91	7.00	5.26	5.13	0.00	
6	9.65	6.66	5.81	6.08	0.00	11.22	6.85	6.52	8.13	0.00	11.21	7.82	8.27	44.57	0.00	12.11	7.51	7.80	6.97	0.00	
7	10.88	7.2	15.91	6.77	0.00	6.44	7.18	6.47	6.14	0.00	9.59	8.48	36.37	7.41	0.00	9.8	22.27	7.73	6.70	0.00	
8	12.37	6.13	7.26	6.76	0.00	9.67	7.40	6.56	6.48	0.00	7.96	6.28	7.16	7.18	0.00	12.5	5.99	6.39	6.25	0.00	
9	21.66	6.00	0.00	0.00	0.00	11.74	6.43	26.20	5.10	0.00	7.24	5.83	6.66	6.23	0.00	8.24	6.15	5.53	5.97	0.00	
10	9.07	6.05	5.78	5.54	0.00	8.74	6.06	6.29	5.58	0.00	8.51	6.29	6.60	21.51	0.00	8.44	5.9	22.47	4.90	0.00	
11	8.73	6.71	5.91	5.87	0.00	9.25	5.64	5.82	5.63	0.00	10.45	6.52	5.97	5.25	0.00	7.94	12.14	6.34	6.13	0.00	
12	10.05	6.24	6.78	6.96	0.00	10.29	7.19	6.63	5.84	0.00	9.74	6.98	7.24	6.31	0.00	12.48	6.58	7.28	7.39	0.00	
13	18.45	6.88	0.00	0.00	0.00	12.29	6.98	7.53	5.74	0.00	8.96	7.54	7.28	9.97	0.00	9.48	7.16	7.12	5.65	0.00	
14	11.12	5.23	13.22	18.48	0.00	9.84	6.50	6.80	5.84	0.00	8.62	6.30	5.88	6.09	0.00	8.63	14.42	6.50	6.33	0.00	
15	9.52	15.65	5.67	6.26	0.00	10.03	6.33	6.75	5.69	0.00	10.46	6.04	6.47	5.94	0.00	18.64	7.29	6.40	5.99	0.00	

[actUeCsirsCoexistCellCsirs = true]

DL BLER	slot_1(	Slot Type																			
	0		1		2		3		4		5		6		7		8		9		
sfn_160ms	D	D	D	S	U	D	D	D	S	U	D	D	D	S	U	D	D	D	S	U	
0	24.89	33.41	6.73	5.85	0.00	35.63	8.89	8.62	6.56	0.00	10.92	18.88	8.19	6.56	0.00	6.91	6.45	6.36	6.34	0.00	
1	7.46	4.70	0.00	0.00	0.00	8.77	6.56	5.96	5.71	0.00	8.37	6.46	6.78	5.84	0.00	9.05	7.83	6.64	5.32	0.00	
3	34.18	5.13	6.61	5.33	0.00	8.94	7.27	7.76	5.37	0.00	7.60	7.13	7.09	5.68	0.00	9.35	8.42	6.52	5.45	0.00	
3	9.63	7.19	7.53	5.93	0.00	9.09	8.27	6.89	6.05	0.00	10.00	6.84	8.30	5.57	0.00	10.06	8.27	8.95	6.28	0.00	
4	41.80	6.52	9.71	5.67	0.00	11.08	8.40	7.58	5.82	0.00	7.36	6.81	6.99	5.10	0.00	9.38	7.80	7.14	5.56	0.00	
5	8.73	6.49	0.00	0.00	0.00	12.15	9.42	8.51	6.61	0.00	11.01	9.45	8.55	7.59	0.00	10.54	10.00	8.69	7.06	0.00	
6	43.74	7.07	8.63	8.09	0.00	8.93	8.52	8.25	6.79	0.00	6.71	6.60	6.67	5.45	0.00	8.28	6.41	5.97	4.37	0.00	
7	8.85	6.61	6.70	4.31	0.00	8.94	7.50	6.59	5.04	0.00	9.32	7.51	6.27	4.95	0.00	9.41	8.59	7.39	5.85	0.00	
8	29.02	6.79	6.86	5.18	0.00	9.56	8.02	8.55	6.51	0.00	7.99	7.10	7.57	4.99	0.00	9.90	7.57	8.11	6.54	0.00	
9	9.18	8.11	0.00	0.00	0.00	11.85	8.76	8.34	6.55	0.00	9.90	9.95	8.43	7.17	0.00	11.19	9.81	7.98	6.97	0.00	
2	34.69	7.13	7.45	5.47	0.00	10.70	7.37	7.97	5.69	0.00	8.64	6.29	7.03	5.39	0.00	8.50	7.02	6.50	5.14	0.00	
11	9.87	7.89	7.43	4.63	0.00	8.67	8.24	6.26	6.06	0.00	10.72	7.15	7.54	5.57	0.00	10.47	8.87	6.99	5.13	0.00	
12	22.75	6.38	8.08	6.26	0.00	10.56	8.51	8.20	6.99	0.00	11.21	8.88	8.44	6.51	0.00	10.82	8.54	8.10	6.60	0.00	
13	9.50	8.04	0.00	0.00	0.00	12.85	10.37	8.80	7.76	0.00	11.19	10.01	9.95	7.32	0.00	9.04	9.67	9.03	7.37	0.00	
1	27.53	7.09	8.86	7.03	0.00	10.22	7.87	7.68	6.10	0.00	9.10	6.47	7.18	5.78	0.00	9.03	6.66	6.70	5.17	0.00	
15	10.23	7.82	7.25	7.12	0.00	8.58	8.88	7.71	5.83	0.00	12.06	8.34	8.41	6.34	0.00	10.50	8.87	7.12	6.24	0.00	

Call summaries		CSIRS offset allocated																													
	CRNTI	23500				29884				50877				14941				60101				43973				33836				716			
	UE CSIRS offset	263				240				102				0				182				280				40				262			
	DL BLER	Slot				Slot				Slot				Slot				Slot				Slot				Slot				Slot			
	SFN mod 16	0	1	2	3	0	1	2	3	0	1	2	3	0	1	2	3	0	1	2	3	0	1	2	3	0	1	2	3	0	1	2	3
	0	29%	26%	5%	7%	51%	27%	6%	8%	36%	23%	7%	8%	15%	28%	7%	9%	49%	24%	5%	8%	23%	24%	6%	14%	44%	33%	6%	11%	49%	32%	6%	6%
	1	11%	6%			12%	7%			8%	8%			6%	5%			12%	8%			10%	6%			7%	5%			10%	7%		
	2	38%	5%	5%	6%	13%	7%	6%	7%	33%	6%	7%	7%	39%	7%	7%	7%	57%	7%	5%	6%	52%	6%	6%	12%	16%	6%	7%	11%	28%	5%	5%	7%
	3	10%	8%	4%	7%	12%	10%	8%	7%	12%	10%	7%	8%	11%	8%	5%	6%	9%	8%	6%	6%	9%	8%	6%	10%	13%	10%	5%	12%	11%	8%	6%	6%
	4	33%	6%	6%	6%	22%	5%	7%	8%	31%	6%	6%	8%	14%	7%	6%	8%	26%	6%	4%	6%	27%	6%	5%	11%	48%	7%	6%	11%	41%	6%	8%	7%
	5	11%	7%			11%	7%			12%	8%			13%	8%			12%	10%			11%	7%			13%	8%			13%	7%		
	6	34%	6%	6%	6%	39%	6%	7%	7%	27%	7%	6%	7%	64%	5%	7%	7%	12%	7%	5%	7%	28%	7%	6%	13%	59%	7%	5%	11%	28%	7%	7%	6%
	7	11%	7%	5%	5%	10%	7%	7%	7%	9%	8%	7%	9%	10%	8%	5%	7%	11%	10%	6%	8%	11%	8%	5%	11%	12%	8%	5%	11%	13%	11%	7%	7%
	8	31%	5%	5%	6%	46%	6%	6%	10%	21%	6%	6%	7%	49%	5%	5%	12%	30%	6%	5%	6%	28%	7%	7%	12%	24%	6%	6%	11%	41%	5%	6%	7%
	9	11%	9%			11%	6%			12%	8%			15%	9%			12%	11%			11%	6%			14%	9%			13%	7%		
	10	24%	6%	5%	5%	61%	7%	6%	7%	27%	7%	8%	7%	28%	7%	7%	7%	47%	8%	6%	8%	32%	8%	10%	15%	59%	7%	7%	13%	59%	5%	5%	6%
	11	13%	7%	6%	5%	11%	11%	7%	7%	11%	9%	7%	7%	12%	9%	5%	8%	10%	9%	6%	7%	11%	9%	7%	13%	11%	9%	6%	11%	10%	9%	5%	6%
	12	30%	6%	8%	8%	15%	7%	6%	8%	14%	5%	5%	6%	40%	7%	6%	7%	27%	7%	6%	8%	26%	7%	6%	12%	42%	7%	6%	12%	30%	6%	6%	10%
	13	11%	7%			12%	8%			11%	8%			12%	9%			12%	9%			12%	7%			15%	8%			13%	8%		
	14	36%	6%	6%	6%	63%	5%	7%	7%	42%	6%	8%	7%	17%	8%	10%	12%	38%	5%	6%	8%	19%	7%	7%	13%	66%	6%	7%	12%	13%	6%	7%	6%
	15	11%	8%	6%	7%	10%	8%	6%	6%	10%	8%	5%	7%	14%	11%	8%	8%	12%	10%	5%	8%	13%	9%	8%	14%	8%	8%	5%	10%	13%	10%	6%	8%

Presented gains shall not be deemed as guaranteed - final figures will always depend on several static or dynamic conditions

Nokia internal use

Technical Details
Knife test results

- Suspecting interference from UE specific CSI-RS of neighbor cells
- W/A: Exclude the last PDSCH symbol of SSB slot from scheduling

Typical transmission on SSB slot : NRB=225, MCS=22, Num layer=3

- With W/A function off, TBS on SSB slot: **524640 bits**
- With function on, TBS on SSB slot: **450984 bits**

Data volume on SSB slots per one second:

- W/A off : 524640 bits x (1000ms/20ms) x 0.6 (BLER 40%) = **15739200 bits**
- W/A on : 450984 bits x (1000ms/20ms) x 0.8 (BLER 20%) = **18039360 bits (14% ↑)**

	Function On		Function Off	
	Slot		Slot	
SFN mod	0	1	0	1
0	9%	19%	20%	20%
1	10%	5%	6%	4%
2	7%	6%	34%	5%
3	11%	9%	10%	7%
4	11%	4%	80%	6%
5	8%	5%	3%	4%
6	10%	7%	73%	4%
7	9%	7%	7%	5%
8	19%	5%	63%	3%
9	4%	4%	6%	4%
10	10%	8%	40%	7%
11	11%	8%	6%	7%
12	10%	5%	8%	6%
13	7%	5%	6%	4%
14	13%	5%	36%	6%
15	8%	5%	8%	6%

CNI-92005 introduces in 23R3 NIDD parameter ***NRBTS.NRCELL.muteLastSymbolInSsbSlot*** enable muting of last symbol in SSB slot
2 options are supported: last symbol can be muted in SSB slot 0 (MutingOfLastDlSymbolInSsbSlotZero) or in all SSB slots)

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Presented gains shall not be deemed as guaranteed - final figures will always depend on several static or dynamic conditions
Nokia internal use

Technical Details

Covariance matrix DTX detection

New improvement

Covariance matrix is **validated** before processing. If received SRS power falls **below threshold** covariance matrix is **not valid.**

Validation of the covariance matrix is done based on the **received SRS power.** If the received SRS power is below a threshold, SRS is considered **not valid.**

- In UL (5GC001086) this threshold is R&D parameter with **a default value of** -132 dBm
  - In UL there is only rank 2 which is handled with 1 beam
  - we can afford to lose fraction of 4RX gain in case bad SRS leads to wrong beam estimation
- In DL (CB008247) in 22R3 2.1PD this threshold is mapped to *NRCELL.**tmpParam3*** (recommended value 1200 corresponding to -120dBm)
  - In DL rank 3 and 4 needs 2 beams, performance is degraded if selected not optimally
  - We need better quality SRS for DL
- In DL, starting from 23R1 release this threshold will be mapped to NIDD parameter *NRCELL.srsForBeamMgmt.**srsPowerThresholdForDlBm*** with **default value** of -120 dBm

Presented gains shall not be deemed as guaranteed - final figures will always depend on several static or dynamic conditions

Nokia internal use

Technical Details

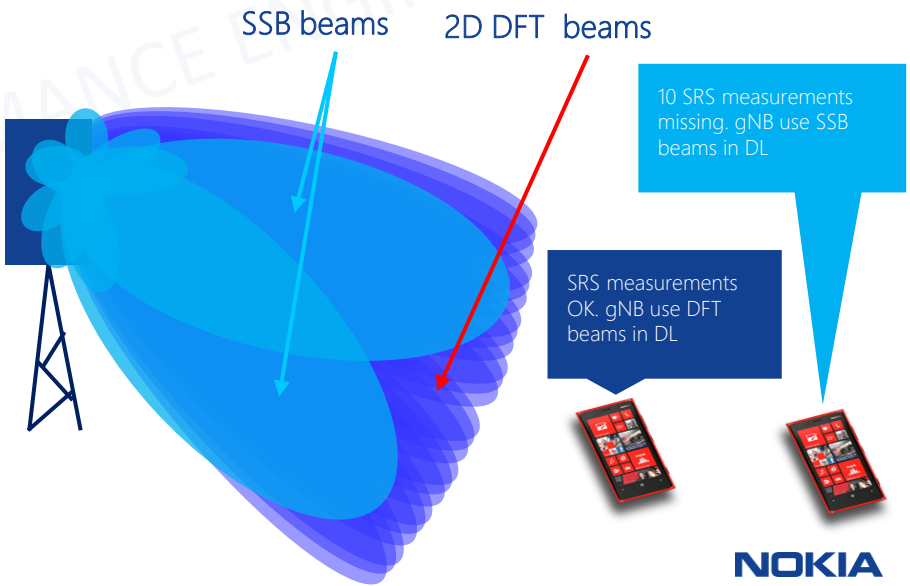
Fallback to legacy beamforming

After a certain time, if **no valid SRS** signal is received, gNB will **fall back to beam reports** from UE. In essence, this **means fallback to the SSB** or refined beams (depends on beam refinement activation, parameter *beamSet.nrBtsBeamRefinementP2*), or **PMI beams** if 5GC002200 is active.

gNB maintains the timer of **covariance matrix age** for **each UE** which is configured with periodic SRS.

If the **age of the covariance** matrix for a given UE **exceeds 10 consecutive SRS** measurements the DL beam(s) are selected based on **beam reports from UE**

gNB will **fall back to UE based reporting** if **10 consecutive SRS** measurements are missed. This is managed by the same logic which is applied to 5GC001086 fallback.



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Presented gains shall not be deemed as guaranteed - final figures will always depend on several static or dynamic conditions

Nokia internal use

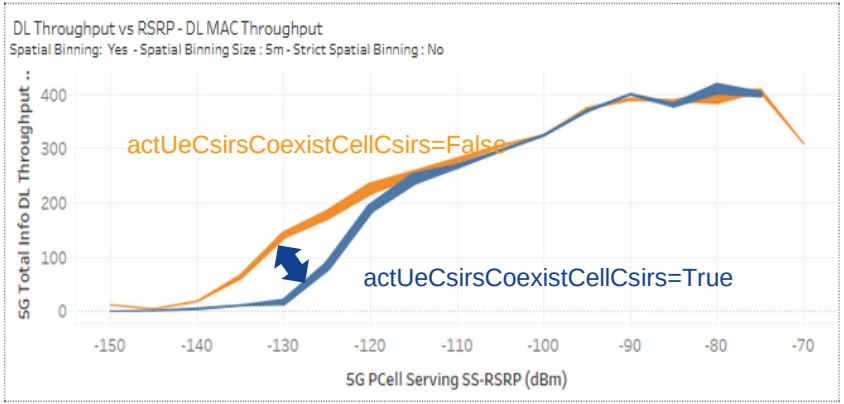
Technical Details
Fallback to legacy beamforming (2)

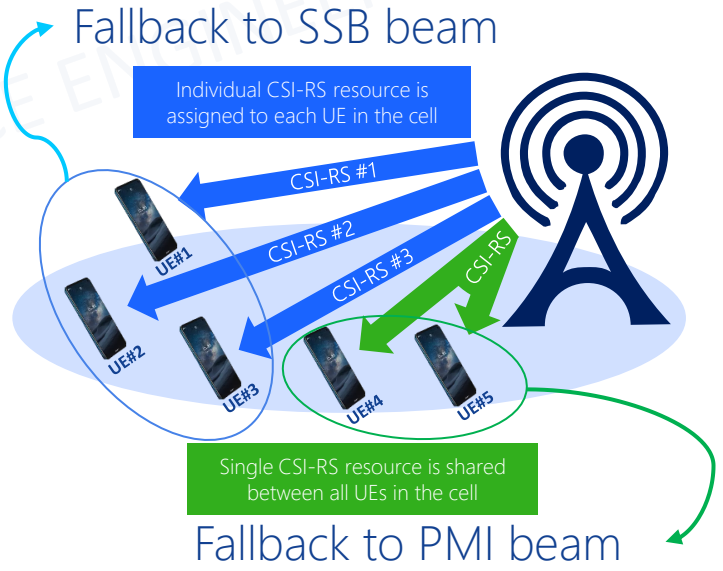
Depending on actUeCsirsCoexistCellCsirs parameter setting and CSI-RS capacity, UE may be configured with UE-specific CSI-RS, or cell specific CSI-RS.

If the **fallback condition is triggered**, following fallback scenario is assumed:

actUeCsirsCoexistCellCsirs=true
UE with UE-specific CSI-RS -> SSB beam
UE with Cell-specific CSI-RS -> PMI beam

As PMI beams perform better than SSB beams, in some scenarios **cell edge UEs** performance may suffer if ***actUeCsirsCoexistCellCsirs=true***





Source: 22R3 BELFORT results Field Performance Report

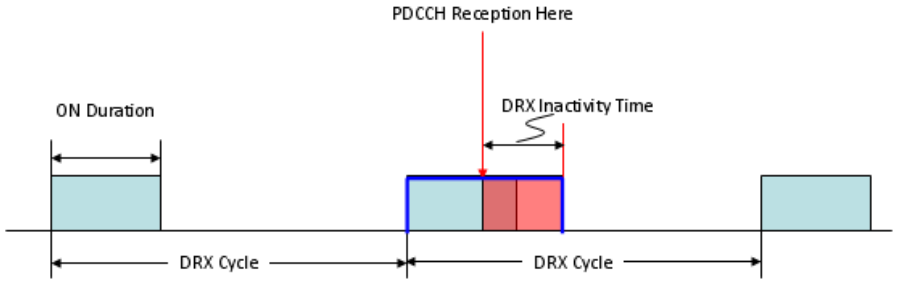
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Presented gains shall not be deemed as guaranteed - final figures will always depend on several static or dynamic conditions
Nokia internal use

Technical Details

Feature Interaction with cDRX

For the interaction between cDRX and SRS allocation, same principles as 5GC001086 are used

- gNB first configures DRX
- After DRX configuration, gNB tries to allocate the SRS in DRX on duration.
- If no free resource is available in DRX on duration, gNB tries to allocate the SRS following DRX on duration and the SRS is as near as possible to DRX on duration. When DRX inactivity timer is running, SRS can be sent.



Presented gains shall not be deemed as guaranteed - final figures will always depend on several static or dynamic conditions
Nokia internal use

Technical Details

CSI-RS Multiplexing

- CB008247 introduces an independent function of **multiple CSI-RS resources multiplexing in slot for both UE and cell specific CSI-RS**.
- The function can be activated and deactivated with the parameter **NRBTS/NRCELL-actCsirsMultiplexing**.
- If **actCsirsMultiplexing** is set to **true**, **2 CSI-RS resources** are **multiplexed in frequency domain** and if the total power exceeds the maximum Tx power, then the bandwidth of CSI-RS for channel acquisition is shrunk to not exeed the maximum Tx power.
- The UE is configured with the number of PRBs satisfying the total power budget.

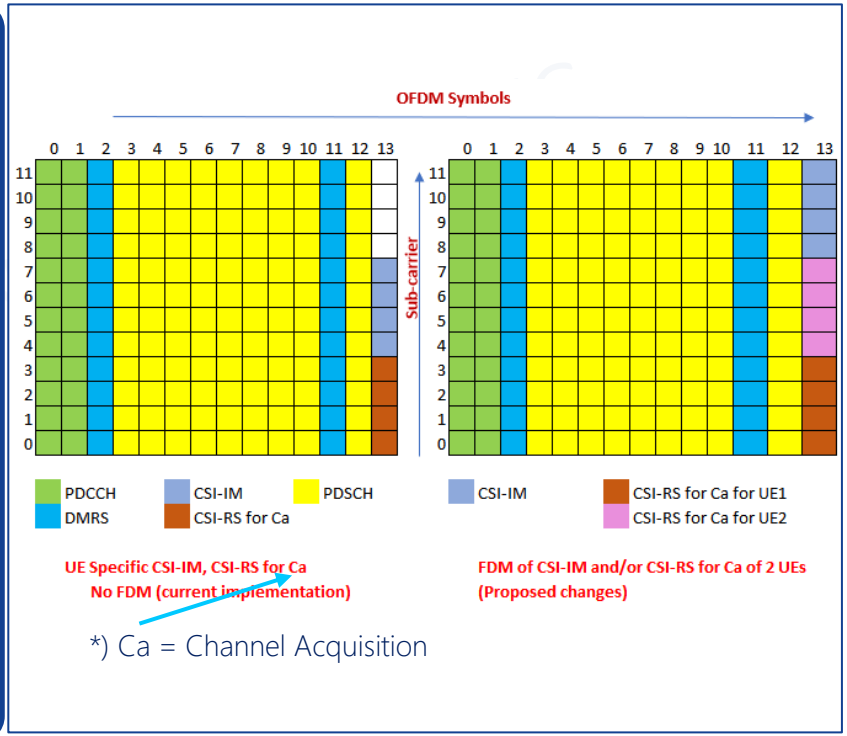
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Technical Details

CSI-RS Multiplexing: UE specific CSI-RS multiplexing

- UE Specific 4 port CSI-RS for Channel Acquisition and UE specific CSI-IM are configured in the same symbols as before. Different CSI-RS resources (for channel acquisition and CSI-IM) are being allocated for different UEs.
- 2 UE specific CSI-RS for Ca resources can be placed in the same symbol of a slot and remaining REs can be used for CSI-IM resources for the two UEs.
- This CSI-IM can be shared between these 2 UEs, which got CSI-RS for Ca scheduled in the same symbol.
- This way, there will be fewer DL symbols (50% less than current implementation) used CSI-RS for Ca. Hence more symbols will be available for PDSCH (over CSI-RS periodicity) and as a result higher DL throughput is expected. More RRC connected UEs will be available with all the same CSI-RS periodicity as before.

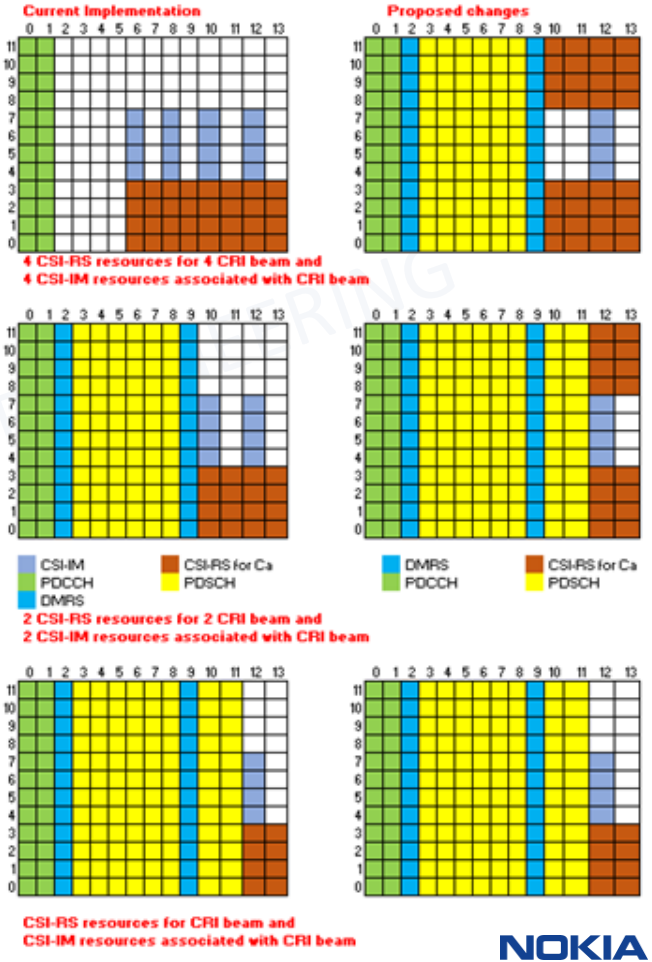


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Technical Details

CSI-RS Multiplexing: Cell specific CSI-RS multiplexing

- FDM of multiple CSI-RS resource(s) belonging to same CSI-RS resourceSet
- Prior introduction of CB008247 or if NIDD:actCsirsMultiplexing is false → gNB keeps different CSI-RS resources of a CSI-RS-ResourceSet in different OFDM symbols.
- After enabling of CB008247 or if NIDD:actCsirsMultiplexing is true → 2 CSI-RS resources will be kept in the same OFDM symbol.
  It will free few OFDM symbols in CSI-RS slots and those can be used for PDSCH.



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Technical Details

CB008247 Improvements: Power Balancing

- For DL SU-MIMO, **each 2 layers** are mapped to **both polarizations** of a **single beam**
- For rank 1 and 2, only the strongest beam is used
- For rank 3 and 4, 2 jointly selected beams are used
- In 5G, there is a **single codeword only** for up 4 layers. Hence, the **same MCS** has to be applied for all of them.
- Especially in case of high coding rates, the overall **performance is dominated by the weakest layer**.

Target of **power balancing** is to **shift power** from the **stronger** layers to the **weaker** ones such that the **optimum performance** is achieved.

Table 7.3.1.3-1: Codeword-to-layer mapping for spatial multiplexing.

Number of layers	Number of codewords	Codeword-to-layer mapping $i = 0,1,\ldots,M_{\text{symb}}^{\text{layer}}-1$	
1	1	$x^{(0)}(i) = d^{(0)}(i)$	$M_{\text{symb}}^{\text{layer}} = M_{\text{symb}}^{(0)}$
2	1	$x^{(0)}(i) = d^{(0)}(2i)$ $x^{(1)}(i) = d^{(0)}(2i+1)$	$M_{\text{symb}}^{\text{layer}} = M_{\text{symb}}^{(0)}/2$
3	1	$x^{(0)}(i) = d^{(0)}(3i)$ $x^{(1)}(i) = d^{(0)}(3i+1)$ $x^{(2)}(i) = d^{(0)}(3i+2)$	$M_{\text{symb}}^{\text{layer}} = M_{\text{symb}}^{(0)}/3$
4	1	$x^{(0)}(i) = d^{(0)}(4i)$ $x^{(1)}(i) = d^{(0)}(4i+1)$ $x^{(2)}(i) = d^{(0)}(4i+2)$ $x^{(3)}(i) = d^{(0)}(4i+3)$	$M_{\text{symb}}^{\text{layer}} = M_{\text{symb}}^{(0)}/4$
5	2	$x^{(0)}(i) = d^{(0)}(2i)$ $x^{(1)}(i) = d^{(0)}(2i+1)$	$M_{\text{symb}}^{\text{layer}} = M_{\text{symb}}^{(0)}/2 = M_{\text{symb}}^{(1)}/3$

Single codeword, up to 4 layers

2 codewords, 5..8 layers

Presented gains shall not be deemed as guaranteed - final figures will always depend on several static or dynamic conditions

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Technical Details

CB008247 Improvements: CNI-84878 (Power Balancing)

Implementation of power balancing related improvement of CB008247 will take place in 22R4 and 23R1

- In 22R4 power balancing is introduced with CNI-84878 only for cell specific CSI-RS and rank 4 only
- Gain is scenario dependent, estimated 0..5% for Rank 4 area
- From SBTS22R4_ENB_0000_000514_000000, functionality is controlled by NRCELL.tmpActFeat1.bit16. Before it was configured with rdBeamPowerBalance R&D parameter
- NRCELL.tmpActFeat1.bit16 is configured as '1' (enable).

CNI-84878 CIS SSD East 1 test result

Rank 4 area: AEQN: 2.08%, AQQL: 2.5%



Whole route

T-put Gain:~1.5%



Rank 4 area

T-put Gain:~2.5%

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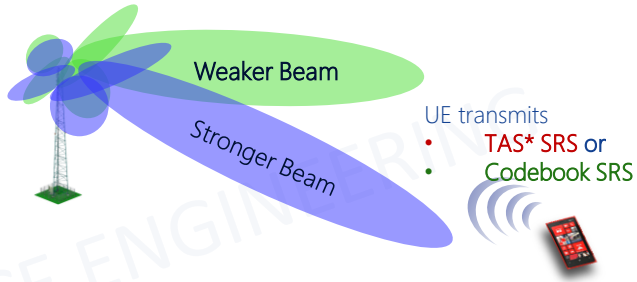
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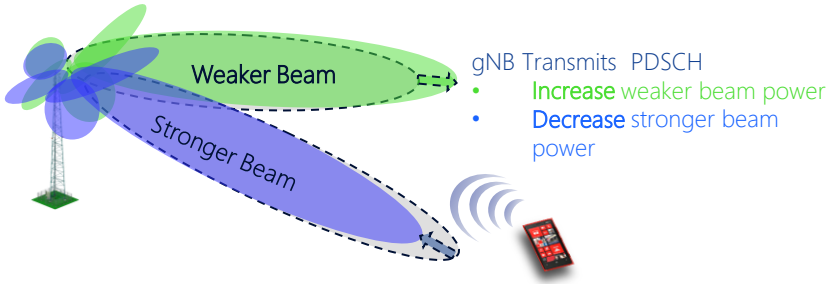
Technical Details

CB010602 SRS based Beamforming extension

In 23R1 power balancing is **enhanced** with CB010602 for

- **Cell specific CSI-RS rank 3** (rank 4 already covered by CNI-84878)
- **UE specific CSI-RS rank 3 and 4**
- Feature gain is estimated 0..2% (based on RRM Feature Gain).
- Gain expectation is extrapolated from CNI-84878 field test result
- Gain is based on statistical samples measurement result, as depending on the radio environment, loss may be observed
- **It is recommended to activate the feature in the cells with high ratio of Rank 4 users.**





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Technical Details

CB010602 SRS based Beamforming extension

CB010602 introduces 2 activation parameters:

- NRCELL.actPdschBeamPowerBalance activates the **PDSCH power balancing** for both cell-specific CSI-RS UEs and **UE-specific** CSI-RS UEs. Parameter **replaces** NRCELL.**tmpActFeat1 Bit 16** parameter used to activate CNI-84878 in 22R4
- NRCELL.actCsirsBeamPowerBalance **activates** the UE specific CSI-RS power balancing . Can only be activated if actPdschBeamPowerBalance is true.
- It is **recommended** to enable actCsirsBeamPowerBalance together with actPdschBeamPowerBalance.

In addition another parameter is introduced:

- NRCELL.beamPowerOffsetThreshold limits the power balancing to the value of this parameter (explained on the next slide).

Next slides explain the parameters in more details.

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Technical Details

CB010602 SRS based Beamforming extension

Following table explains the impact of SRS beam power balancing related parameters
Precondition: Cell-specific CSI-RS **enabled** (dlMimoMode = **8x4 Closed Loop Spatial Multiplexing**).
Otherwise, **cell-specific CSI-RS is not enabled** and UEs are only configured with UE-specific CSI-RS.
Power balancing may **still apply** to UEs configured with UE-specific CSI-RS.

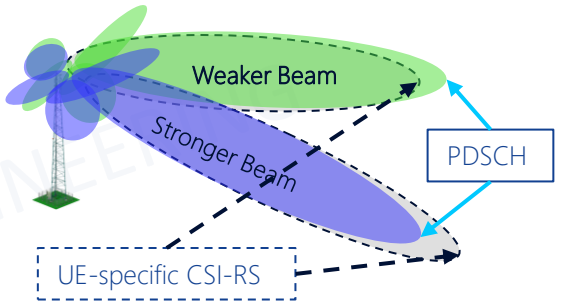
actUeCsirsCoexistCellCsirs = false

parameter	UEs with **UE specific CSI-RS**	UEs with **cell specific CSI-RS**
actPdschBeamPowerBalance = false actCsirsBeamPowerBalance = n/a	Not configured	PDSCH power balancing disabled
actPdschBeamPowerBalance = true actCsirsBeamPowerBalance = n/a	Not configured	PDSCH power balancing active

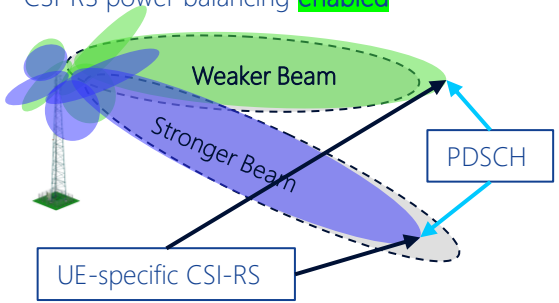
actUeCsirsCoexistCellCsirs = true

parameter	UEs with UE specific CSI-RS	UEs with cell specific CSI-RS
actPdschBeamPowerBalance = true actCsirsBeamPowerBalance = false	PDSCH power balancing active CSI-RS power balancing disabled	PDSCH power balancing active
actPdschBeamPowerBalance = true actCsirsBeamPowerBalance = true	PDSCH power balancing active CSI-RS power balancing enabled	PDSCH power balancing active

Example:
PDSCH power balancing active
CSI-RS power balancing disabled



Example:
PDSCH power balancing active
CSI-RS power balancing enabled



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Technical Details

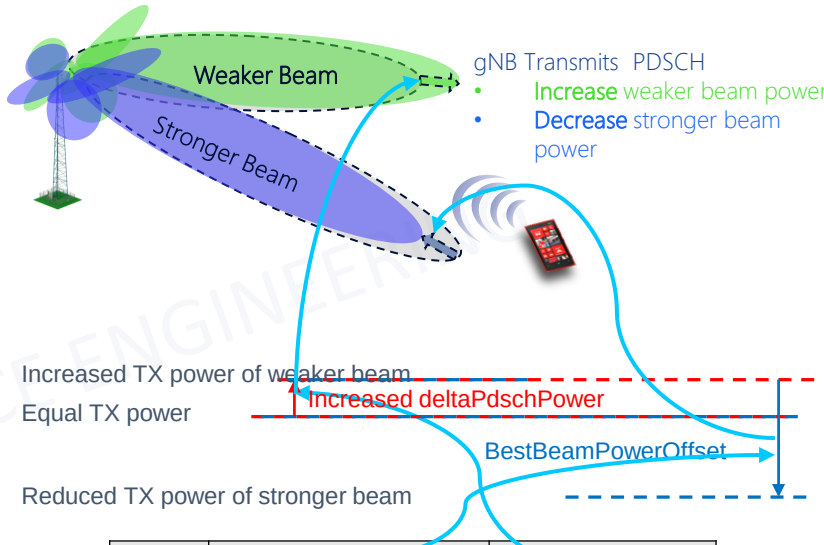
CB010602 SRS based Beamforming extension

How power balancing works?

- Beam power is measured on the received SRS symbols
- Beam ratio is calculated and converted in dB scale:

$$r_{dB} = 10log\left(\frac{P_{weaker}}{P_{stronger}}\right)$$

- Next larger entry in **bestBeamPowerOffset** column than r_{dB} from the look-up table is selected.
- **Weaker beam** is **increased** by **deltaPdschPower** dBs while **stronger beam** is **decreased** by **bestBeamPowerOffset** dBs
- There are 2 different tables for rank 3 and rank 4 to account for different numer of polarities per layer
- beamPowerOffsetThreshold **limits** the **power balancing to** the value of this parameter (default value 6dB)



Index	bestBeamPowerOffset [dB]	deltaPdschPower [dB]
0	0	0
1	-1	0,4
2	-2	0,8
3	-3	1,2
4	-4	1,5
5	-5	1,8
6	-6	2
7	-8	2,3
8	-10	2,5
9	-15	2,8

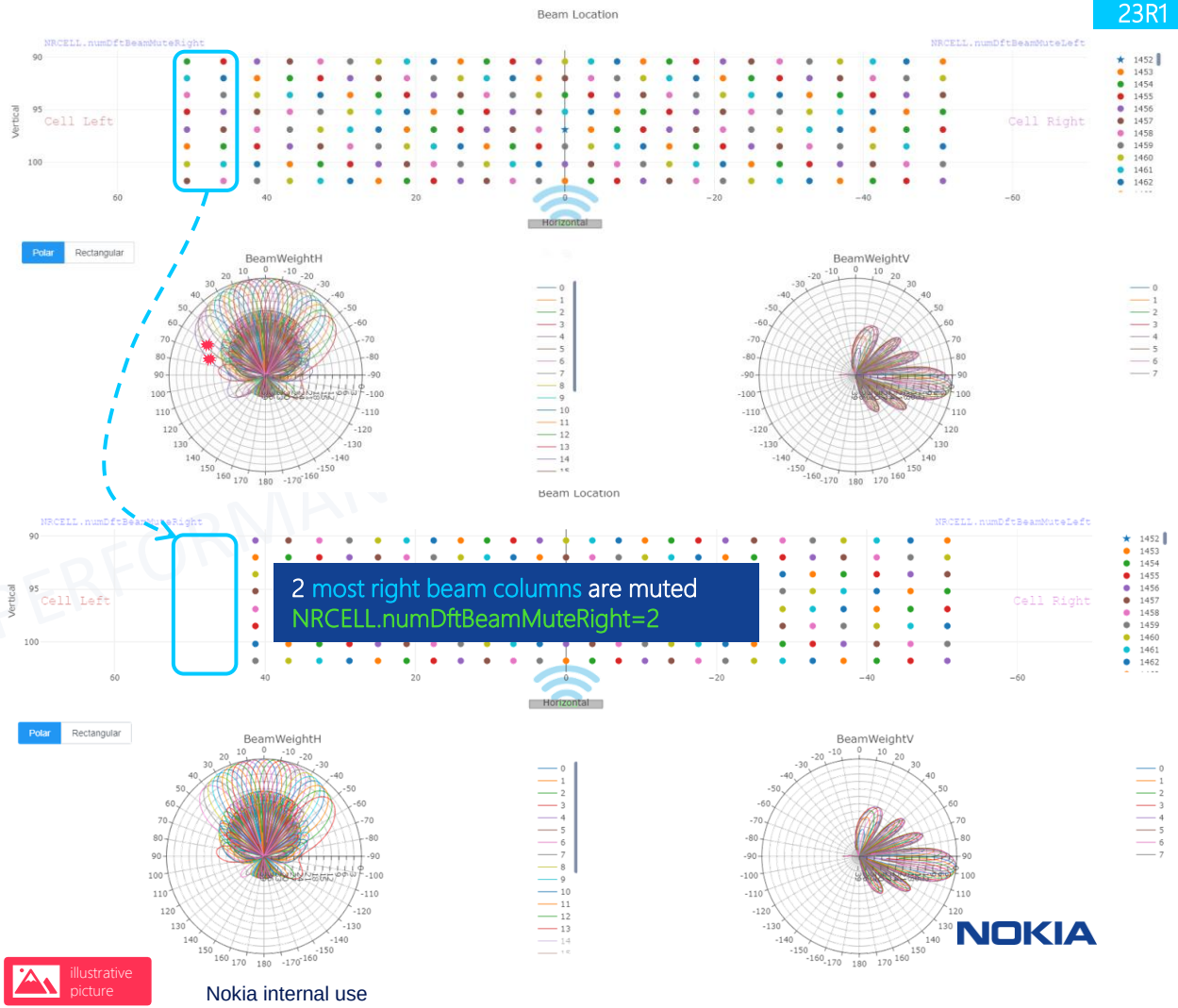
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Technical Details
2D DFT beams muting

- In 23R1 SW release there are 2 new NIDD parameters introduced:
  - NRCELL.numDftBeamMuteRight
  - NRCELL.numDftBeamMuteLeft
- Those parameter indicates **the number of right/left most DFT beams in horizontal plane which are muted**. The number of beams muted is relative to the full DFT beam set. Value 0 means the full DFT beam set in right/left side is in use. Value 1 means 1 beam relative to the full DFT beam set in right/left side is muted.
- If this parameter is not configured, then the DFT beams in right/left side **are internally determined by gNB according to the opening angle setting**.

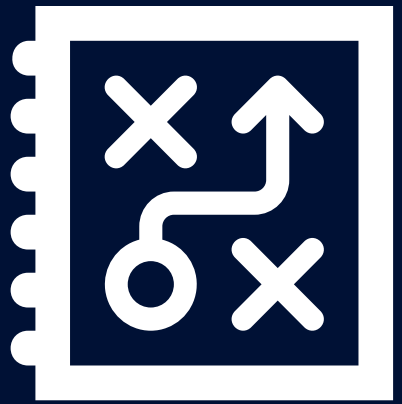
New beam muting parameters **extend flexibility of the 2D DFT beams configuration** by possibility of independed muting of 2D DFT beams on left and right side of cell (looking from UE perspective).



illustrative picture

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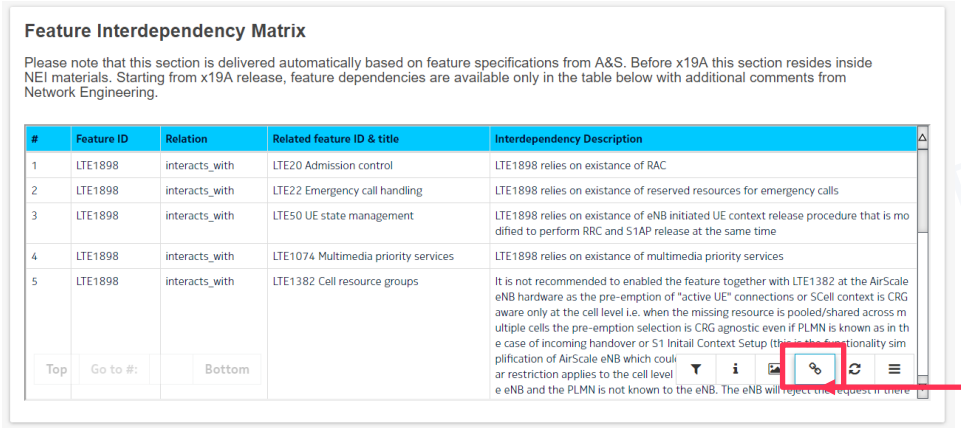
Interdependencies



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How to find feature interdependencies in WebNEI?

- Under Network Engineering Information materials for certain feature there is dedicated section called Feature Interdependency Matrix, which describes feature relations based on R&D input with additional NetEng add-ons:



Feature Interdependency Matrix

Please note that this section is delivered automatically based on feature specifications from A&S. Before x19A this section resides inside NEI materials. Starting from x19A release, feature dependencies are available only in the table below with additional comments from Network Engineering.

#	Feature ID	Relation	Related feature ID & title	Interdependency Description
1	LTE1898	interacts_with	LTE20 Admission control	LTE1898 relies on existance of RAC
2	LTE1898	interacts_with	LTE22 Emergency call handling	LTE1898 relies on existance of reserved resources for emergency calls
3	LTE1898	interacts_with	LTE50 UE state management	LTE1898 relies on existance of eNB initiated UE context release procedure that is mo dified to perform RRC and S1AP release at the same time
4	LTE1898	interacts_with	LTE1074 Multimedia priority services	LTE1898 relies on existance of multimedia priority services
5	LTE1898	interacts_with	LTE1382 Cell resource groups	It is not recommended to enabled the feature together with LTE1382 at the AirScale eNB hardware as the pre-emption of "active UE" connections or SCell context is CRG aware only at the cell level i.e. when the missing resource is pooled/shared across m ultiple cells the pre-emption selection is CRG agnostic even if PLMN is known as in th e case of incoming handover or S1 Initail Context Setup (th plification of AirScale eNB which coul ar restriction applies to the cell level e eNB and the PLMN is not known to the eNB. The eNB will

Top Go to #: Bottom

- If necessary, you can always download a copy of the relations by pushing a button in right corner
- Feature relations which might impact network performance (positively or negatively) are described in more details at next slides. Those relations are also marked with special icon on our Feature Interdependency Matrix module (add new screenshot when new view will be ready).

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Feature dependencies

Feature 5GC002516 Coordination of TDD deployments over country borders interacts with features 5GC001086 and CB008247

Capacity and Performance impact		
	DL	UL
Throughput	()	YES (S) (▼)
Coverage	()	()
Max # of RRC Conn users/cell	()	
Others	()	

State for 7.12.2021 (theoretical analysis)

- **UL Throughput**: Special slot will not be used for periodic SRS resource allocation if it is mapping to a DL slot in the interferering cell.
- This will lead to less opportunities for periodic SRS and in turn less UEs scheduled with SRS based beamforming

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<5G_FIM_12744>

Feature dependencies

Feature CB008247 Interact with feature 5GC002052 EIRP control

Capacity and Performance impact		
	DL	UL
Throughput	YES (S) (▼)	NO
Coverage	YES (S) (▼)	NO
Max # of RRC Conn users/cell	NO	
Others	NO	

- Throughput: Power decrease with EIRP control on selected beams
- Coverage: Power decrease with EIRP control on selected beams

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<5G_FIM_12745>

Feature dependencies

Feature CB008247 Interact with feature 5GC002200 support for 4x8 port CSI-RS for SU-MIMO

Capacity and Performance impact		
	DL	UL
Throughput	NO	NO
Coverage	NO	NO
Max # of RRC Conn users/cell	YES (I) (▲)	
Others	()	

- Max number of RRC connected UEs is increased due to cell specific CSI-RS in use, which allows to connect more users than UE specific CSI-RS

Please note: 5GC002200 activation is not mandatory with CB008247

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Performance Executive Summary



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CB008247 Performance Executive Summary

Feature monitoring	
New counters	NO
Related legacy counters	YES
KPIs	YES
Activation Monitoring via CM	YES

Capacity and Performance impact	DL	UL
User Peak Rate	NO	NO
Cell Throughput	▲ YES(I)	NO
Cell Edge User Throughput	NO	NO
Maximum Cell Range	NO	NO
Maximum # of RRC Conn users/cell	▲▼ YES(S)	
E2E Delay	NO	NO
Accessibility	NO	
Retainability	NO	
Power Consumption	▼ YES(S)	
Others	NO	

Factors impacting feature performance	
Dependency on configuration	RU specific - gain depends on DFT size which in turn depends on RU array dimensions/ beamset configuration. Long PUCCH only, ABIO/ASOE, valid nrCellType
Other dependency	Share of UEs with SRS antenna switching capabilites, number of UEs, UE distribution, legacy beamset
Comment	Average DL cell TP is expected to increase after feature activation; 2D DFT GoB size and resolution depends on the number of TRXs and beamset opening angle; in case of higher # of UEs some SRS cannot be processed due to L1 limitation; vertical coverage improvement in near cell

Comments to impacted areas

- Cell TP: 2D DFT beamset is used in DL - higher number of beams provides better beam resolution
- Max # conn. UEs: Since CSI-RS multiplexing can be enabled regardless of whether "SRS Based DL BF" is activated or not, # of RRC Connected UEs can be increased for the cases where UE specific CSI-RS has previously been the limiting factor.
- Power Consumption: UE power consumption increased due to sending of periodic SRS - theorethical assumption; no impact on gNB power consumption

GREEN - positive; **RED** - negative; **ORANGE** - depends on scenario; **GRAY** - no impact

YES (I) (▲) - there is intended **positive** impact of the feature (by design)
YES (I) (▲▼) - impact of the feature **depends on the scenario** and is intended (by design)
YES (S) (▲) - there is **positive** impact of the feature which is a side effect (not intended)
YES (S) (▼) - there is **negative** impact of the feature which is a side effect (not intended)
YES (S) (▲▼) - impact of the feature depends on the scenario and is a side effect (not intended)
NO - there is no impact

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CB008247 Performance Executive Summary

Gain impacting factors

- Achievable feature gains depend on many factors:
  - RU antenna array architecture – with different RU array layout different size of DFT beamset is possible, the more beams, the higher the beamset resolution and performance gain
  - Legacy SSB beamset – different legacy beamsets will have varying performance, which influences overall feature gain
  - Legacy beam refinement - legacy beam refinement already provides some gain which influences overall CB008247 gain
  - 5GC002200 activation – which already provides some gains with respect to SSB based beamforming
  - Number of UEs – due to HW limitation some UEs SRS will not be processed; these UEs will be scheduled with legacy beamsets. This is relevant to cells with very high numer of connected UEs
  - Number of UEs with SRS TAS capability – since this capability is optional, some UEs will be using codebook SRS wshich may impact performance

NETWORK PERFORMANCE ENGINEERING

CB008247 Performance Executive Summary

Optimal and Problematic Scenarios for CB008247

Most optimal scenarios for DL SRS:

- Thanks to larger vertical steering range DL SRS gives good gains when UE is **out of legacy BF feature vertical coverage**
- Thanks to better beamforming gain DL SRS provides better throughput in **Rank 2 area**

Most problematic scenarios for DL SRS:

- **Line Of Sight** scenario that is within 5GC002200 coverage area or 5GC002200 has good reflective buildings behind creating multipath

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Expected feature gains

FR1 TDD – Cell throughput gains

P&C requirements | Simulation results | NRdim Tool calculations | Theoretical calculations | Test results

Avg. cell throughput

Source: RRM Feature Gains

Baseline common configuration	Feature specific configuration	Cell specific configuration	Maximum gain forecast
• TDD 100MHz @ 3.5Gz, single carrier • BS height: 25m, mech. downtilt 6 deg • 21 cells, UMa no buildings, LoS + NLoS • ISD500 • Full buffer traffic • Antenna model: 192AE 64T64R, 200W • UL/DL ratio 5GC001208 DL:UL=4:1 • CQI period: 160ms • SU-MIMO only • UE speed: 3km/h • UE receiver: 4RX IRC, 256QAM	DL BF algorithm: • Baseline: 5GC002200 cel specific 8 port CSI-RS 2x2 split 4 resources • Feature on: up to 2 best beam based on pSRS measurement, phase offset btwn. 2 polarizations based on UE feedback i2	Single UE single cell	Tbd
		Average MU DL Throughput	4.5%..9%

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Source: Classic_gNB_Capacity_Req_FR1TDD

Expected gains

FR1 TDD – RRC Connected UE

P&C requirements | Simulation results | NRdim Tool calculations | Theoretical calculations | Test results

RRC Connected UEs per cell Note 1

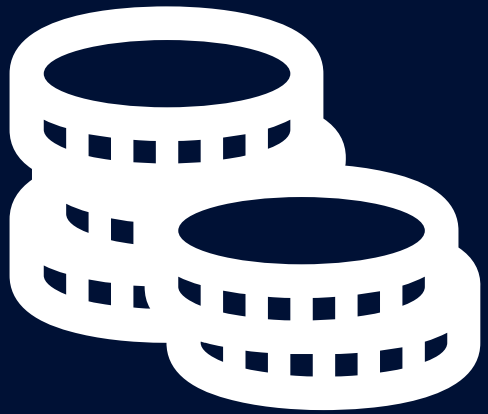
Source: Classic_gNB_Capacity_Requirements_FR1TDD

Baseline common configuration	Feature specific configuration	Cell specific configuration		Max # of UEs	Max # of UEs with DL SRS BF
• Cell BW: 100MHz • dlMimoMode 8x4 • **NOTE 1:** Numbers are UEs which can use SRS BF. The remaining RRC Connected UEs will use legacy BF and are still allowed to enter the cell.	• # of cells in ABIO/ASOE 6 • numberOfTransmittedSsBlocks 6 (1208/1070) • numberOfTransmittedSsBlocks 8 (1116_3ms) • numberOfTransmittedSsBlocks 4 (1116_0ms) • ssBurstSetPeriod (ms) 20 • csirsTrackingPeriod (ms) 80 • csiReportPeriodicity slots320 • pucchModeSelect longPucch • actFdmEnhancedScheduling TRUE • numberOfRxBeamforming 4 • actUlSrsBm TRUE • actdlSrsBm TRUE • srsUePeriodicity slots40	supportedSRS-TxPortSwitch == t1r1 (or t2r2 or t4r4)	TDD Frame structure = 5GC001208	512 (NSA) 480 (SA)	512
			TDD Frame structure = 5GC001116_0ms/3ms	704 (NSA) 672 (SA)	512
			TDD Frame structure = 5GC001070	448 (NSA) 768 (SA)	544
		supportedSRS-TxPortSwitch == t1r2 (or t2r4)	TDD Frame structure = 5GC001208	512 (NSA) 480 (SA)	362
			TDD Frame structure = 5GC001116_0ms/3ms	704 (NSA) 672 (SA)	362
			TDD Frame structure = 5GC001070	448 (NSA) 416 (SA)	362
		supportedSRS-TxPortSwitch == t1r4	TDD Frame structure = 5GC001208	512 (NSA) 480 (SA)	217
			TDD Frame structure = 5GC001116_0ms/3ms	704 (NSA) 672 (SA)	217
			TDD Frame structure = 5GC001070	448 (NSA) 416 (SA)	217

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Expected Benefits & Performance Impact



Presented gains shall not be deemed as guaranteed - final figures will always depend on several static or dynamic conditions
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Coverage aspects

- Feature introduces new 2D DFT beamset for DL which should outperform the legacy beamset using refined beams and PMI beams
- Due to **denser grid** of beams **vertical cell coverage** should be improved overall, especially bringing benefit to UEs in near cell
- Flexibility of 2nd beam selection will improve DL throughput compared to PMI beams
- Maximum cell range is **not expected to be impacted** since gNB will fall back to legacy beams once SRS signal becomes unreliable

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Radio Capacity aspects
Simulation Limitation

In MoRSE, no link level procedure, thus assume **all control channel work perfectly**.

- gNB **always received CQI** report correctly.
- gNB **always detected the SRS** signal.
- **No channel estimation error** modeling.



Due to those limitations, below behavior <u>will not happened</u> in MoRSE:

- **Beam fall back** due to no valid report or measurement.
- **Too weak signal** cause large channel estimation error, finally cause **wrong beam selection**.

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Radio Capacity aspects

Simulation configurations – Single Cell

deployment scenarios	carrier frequency and bandwidth	TDD 100MHz @ 3.5GHz, single carrier only
	site/cell configuration	BS height = 25m; mechanical down-tilt: 6 degree
		1 cell , no buildings, LoS and NLoS mixed
		ISD500
	Traffic model	full buffer
	Channel model	Uma without building from TR 38.901
gNB configuration	antenna model	192AE->4x8x2@ 64T64R , output 200W
	TDD subframe configuration	DL:UL = 4:1, aligned up to 5GC001208
	CQI Measurement	160ms
	CQI report period	160ms
	BLER target	10%
	downlink BF algorithm	
	Baseline (5GC2200)	Cell specific CSI-RS with 8 ports: 2x2 split #4 resource; 4x1 split #4 resource
		precoding = 3GPP codebook; beamforming = CSI-RS sub-panel beam
	Option-1	Cell specific CSI-RS with 8 ports: 2x2 split #4 resource; 4x1 split #4 resource
		Up to 2 best beam based on pSRS measurement. Phase offset between two polarizations based on UE feedback i2.
	downlink MU MIMO	disabled
	UL/DL scheduler	RRM SFS scheduler: max 16 scheduled users per TTI, with eCPRI sub-cell
	Link Adaptation	5GC001796 LA
UE configuration	downlink UE receiver	4Rx IRC receiver, 256QAM supported
	UE movement	3km/h, Multiple session to cover whole cell area

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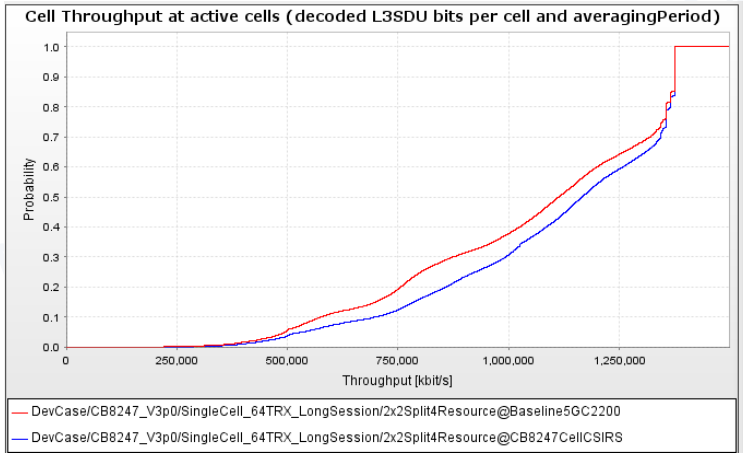
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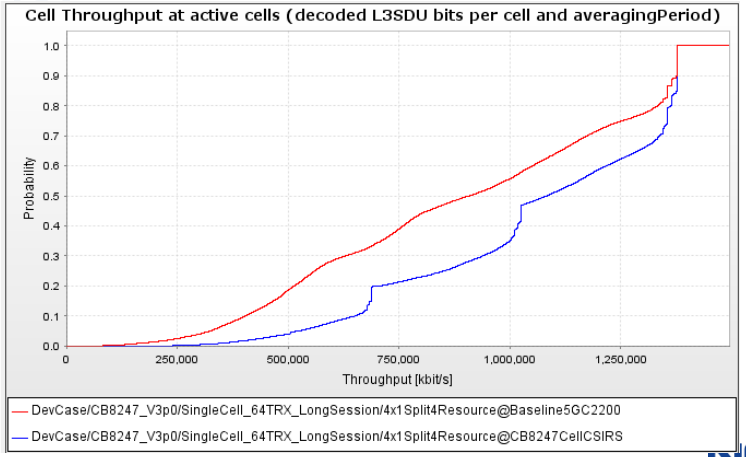
Radio Capacity aspects

Single Cell Single UE – Cell Throughput

Cell specific CSI-RS

	5GC2200			CB008247			Gain		
	5%	Mean	95%	5%	Mean	95%	5%	Mean	95%
2x2 split #4 resource	0.49 Gbps	1.11 Gbps	1.37 Gbps	0.53 Gbps	1.16 Gbps	1.37 Gbps	8.1%	4.5%	0%
4x1 split #4 resource	0.32 Gbps	0.91 Gbps	1.37 Gbps	0.52 Gbps	1.08 Gbps	1.37 Gbps	62.5%	18.6%	0%





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Radio Capacity aspects

Simulation configurations – 21 Cell 210 Users

deployment scenarios	carrier frequency and bandwidth	TDD 100MHz @ 3.5GHz, 21 cells
	site/cell configuration	BS height = 25m; mechanical down-tilt: 6 degree
		21 cell , no buildings, LoS and NLoS mixed
		ISD500
	Traffic model	full buffer
	Channel model	Uma without building from TR 38.901
gNB configuration	antenna model	192AE->4x8x2@ 64T64R , output 200W
	TDD subframe configuration	DL:UL = 4:1, aligned up to 5GC001208
	CQI Measurement	160ms
	CQI report period	160ms
	BLER target	10%
	downlink BF algorithm	
	Baseline (5GC2200)	Cell specific CSI-RS with 8 ports: 2x2 split #4 resource; 4x1 split #4 resource
		precoding = 3GPP codebook; beamforming = CSI-RS sub-panel beam
	Option-1	Cell specific CSI-RS with 8 ports: 2x2 split #4 resource; 4x1 split #4 resource
		Up to 2 best beam based on pSRS measurement. Phase offset between two polarizations based on UE feedback i2.
	downlink MU MIMO	disabled
	UL/DL scheduler	RRM SFS scheduler: max 16 scheduled users per TTI, with eCPRI sub-cell
	Link Adaptation	5GC001796 LA
UE configuration	downlink UE receiver	4Rx IRC receiver, 256QAM supported
	UE movement	3km/h

NETWORK PERFORMANCE ENGINEERING

Presented gains shall not be deemed as guaranteed - final figures will always depend on several static or dynamic conditions

Nokia internal use

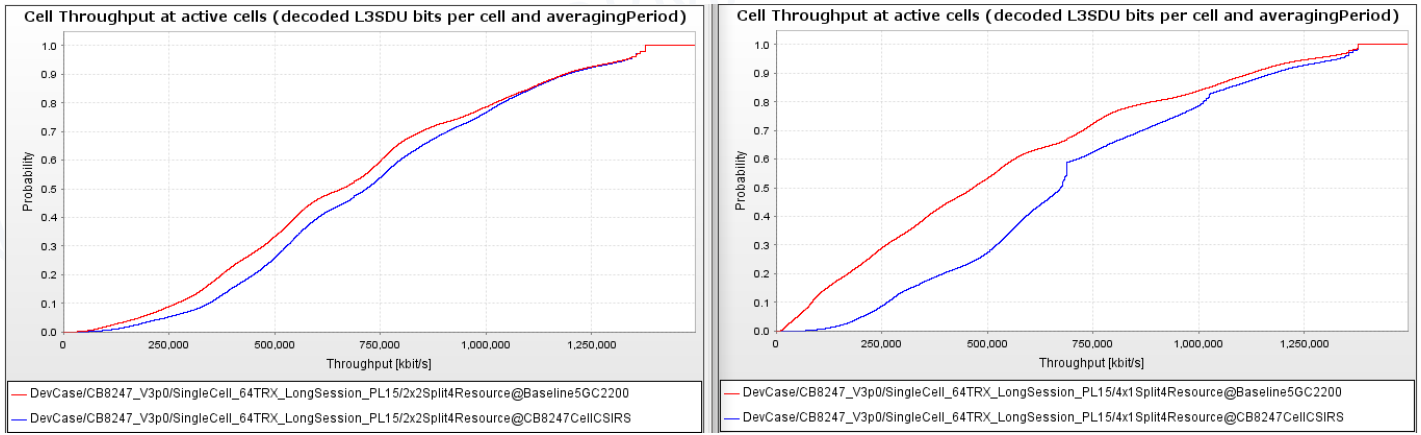
Radio Capacity aspects

Single Cell Single UE – Cell Throughput

Cell specific CSI-RS

	5GC2200			CB008247			Gain		
	5%	Mean	95%	5%	Mean	95%	5%*	Mean	95%
2x2 split #4 resource	0.18 Gbps	0.66 Gbps	1.32 Gbps	0.24 Gbps	0.72 Gbps	1.33 Gbps	33.3%	9.1%	0.7%
4x1 split #4 resource	0.05 Gbps	0.47 Gbps	1.27 Gbps	0.20 Gbps	0.67 Gbps	1.36 Gbps	400%	42.5%	7.1%

* With additional path loss, the 5% point maybe too extreme to be realistic. Shows it just to indicate the trend



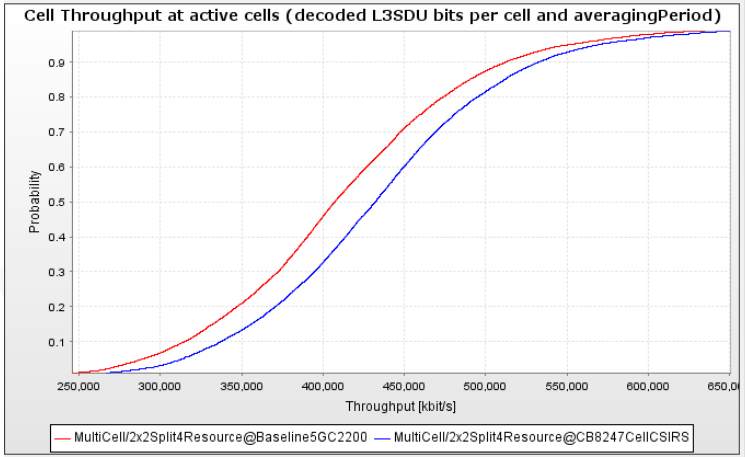
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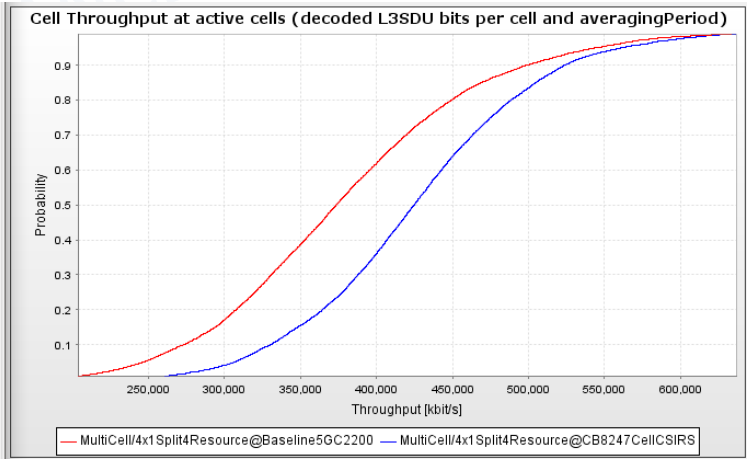
Nokia internal use

Radio Capacity aspects

Multiple Cell Multiple UE – Cell Throughput

	5GC2200			CB008247			Gain		
	5%	Mean	95%	5%	Mean	95%	5%	Mean	95%
2x2 split #4 resource	0.29 Gbps	0.41 Gbps	0.55 Gbps	0.31 Gbps	0.43 Gbps	0.57 Gbps	6.8%	4.8%	3.6%
4x1 split #4 resource	0.25 Gbps	0.37 Gbps	0.54 Gbps	0.31 Gbps	0.42 Gbps	0.56 Gbps	24.0%	13.5%	3.7%





Presented gains shall not be deemed as guaranteed - final figures will always depend on several static or dynamic conditions

Nokia internal use

Specific settings found in one of customers network

Radio Capacity aspects

Simulation Configuration and Scenario

deployment scenarios	carrier frequency and bandwidth	TDD 100MHz @ 3.5GHz, single carrier only
	site/cell configuration	BS height = 25m; mechanical down-tilt: 6 degree
		1 cell , no buildings, LoS and NLoS mixed
		ISD500
	Traffic model	full buffer
	Channel model	Uma without building from TR 38.901
gNB configuration	antenna model	128AE->2x8x2@ 32TRX (**AEQY**); 96AE->2x8x2@ 32TRX (**AEQN**)
	TDD subframe configuration	DL:UL = 4:1, aligned up to 5GC001208
	CQI Measurement	160ms
	CQI report period	160ms
	BLER target	10%
	downlink BF algorithm	
	Baseline (5GC2200)	2x2 split #4 resource; 4x1 split #1 resource; Custom 4x1 split #2#2 resource
	CB8247	CSI-RS same with 5GC2200. PDSCH using oversampled DFT beams.
	downlink MU MIMO	disabled
	UL/DL scheduler	RRM SFS scheduler: max 16 scheduled users per TTI, with eCPRI sub-cell
	Link Adaptation	5GC001796 LA
UE configuration	downlink UE receiver	4Rx IRC receiver, 256QAM supported
	UE movement	**3km/h**, Multiple session to cover whole cell area **30km/h**, Multiple session randomly walk in cell area

ecific CSI-RS
32 TRX RU

See https://webnei.emea.nsn-net.net/#/webnei/5fb65eb9a345c2001ebb9cf8/164 for custom #2#2 resource configuration

NETWORK PERFORMANCE ENGINEERING

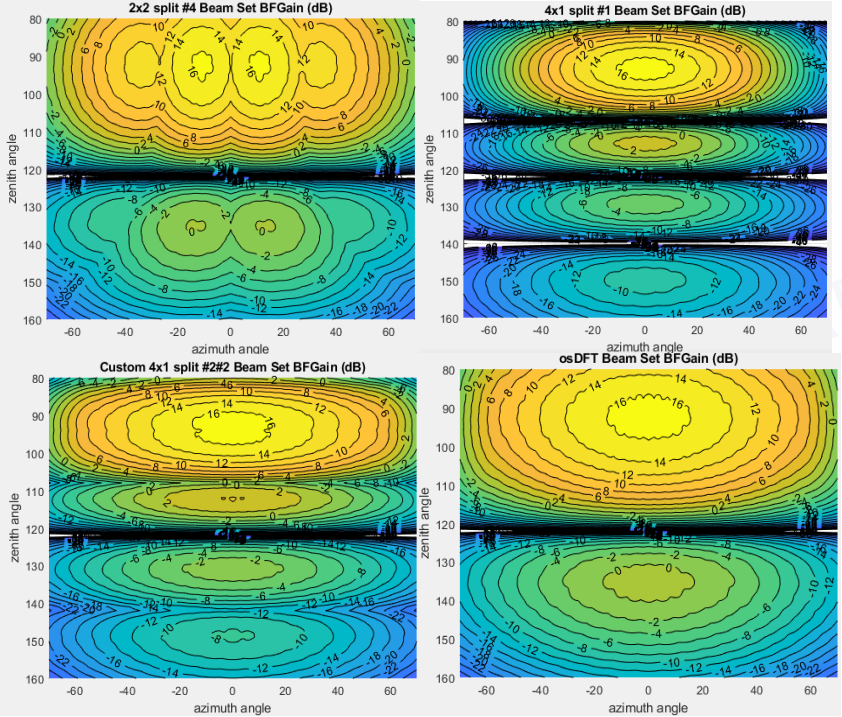
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Nokia internal use

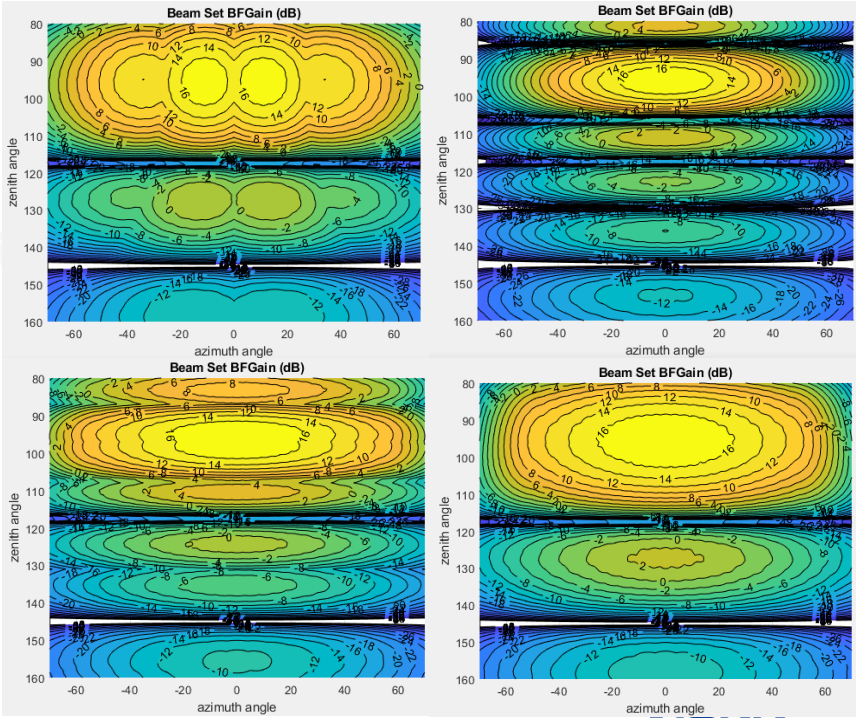
Radio Capacity aspects

Beamset analysis

AEQN: 3AE/TRX, pre-tilt 3 deg



AEQY: 4AE/TRX, pre-tilt 6 deg



Presented gains shall not be deemed as guaranteed - final figures will always depend on several static or dynamic conditions

Nokia internal use

Radio Capacity aspects

Throughput Analysis

Cell specific CSI-RS
32 TRX RU

- Table shows **median DL TP gains** for different combinations of cell specific CSI-RS configurations, UE speed and RUs.
- Best gains are seen with standard 4x1 CSI-RS configuration.

UE speed	AEQN			AEQY		
	2x2 #4	4x1 #1	4x1 customized	2x2 #4	4x1 #1	4x1 customized
3km/h	0.97%	4.14%	0.86%	0.95%	13.77%	6.78%
30 km/h	0.99%	5.14%	0.1%	0.79%	14.12%	5.47%

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Presented gains shall not be deemed as guaranteed - final figures will always depend on several static or dynamic conditions

Nokia internal use

Observed Performance Results



Presented gains shall not be deemed as guaranteed - final figures will always depend on several static or dynamic conditions
Nokia internal use

Observed Performance Results

Feature Corner – Performance Hub

Feature Corner – link

5GC001086 feature performance reports can be found in 22R2 Performance Knowledge Hub – link
CB008247 feature performance reports can be found in 22R3 Performance Knowledge Hub – link

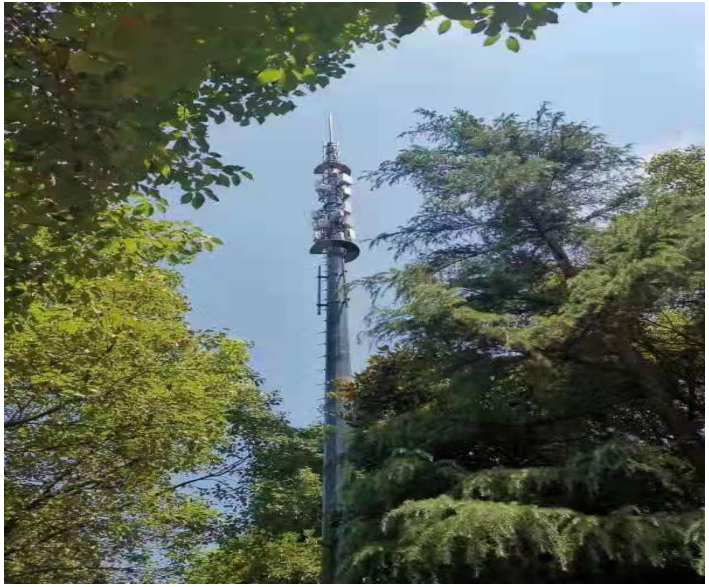
NETWORK & PERFORMANCE ENGINEERING

Observed Performance Results

CIS SSD EAST field results

Source: https://nokia.sharepoint.com/:p:/r/sites/MNRITSSDEAST/Shared%20Documents/50%20Biz%20Program/CIS_SSD_East1_FTN/07.PoC_Demo_TF/8247/CB8247%20Option3.2%20result%20.pptx?d=w3ac215c57bee471b9dc9a322ea6b2504&csf=1&web=1&e=qelmD4

Test was performed in rural area including LOS and NLOS test route



Key NIDD Parameter settings

- 1070(3:7) frame structure
- BW 100MHz
- Beamset_6
- PRACH Index 191
- 256QAM
- CL8x4 MIMO
- pMax 33.9dBm
- ssPBCH 12 dBm
- 256QAM power backoff 3
- 8X4 MIMO （2200）
- NRBTS/NRCELL-actUlSrsBm = true
- NRBTS/NRCELL-actDlSrsBm = true
- actCsirsMultiplexing = False
- actPdschAtSsbSlots = False
- actPdschRmCsirsForTracking = False

SiteName	PCI	Radio	BandWidth	Amuzth	Ant. Height	Down Tilt
DX_6	195	AEQB	100MHz	45	27	0
	196	AEQB	100MHz	135	27	0

Presented gains shall not be deemed as guaranteed - final figures will always depend on several static or dynamic conditions

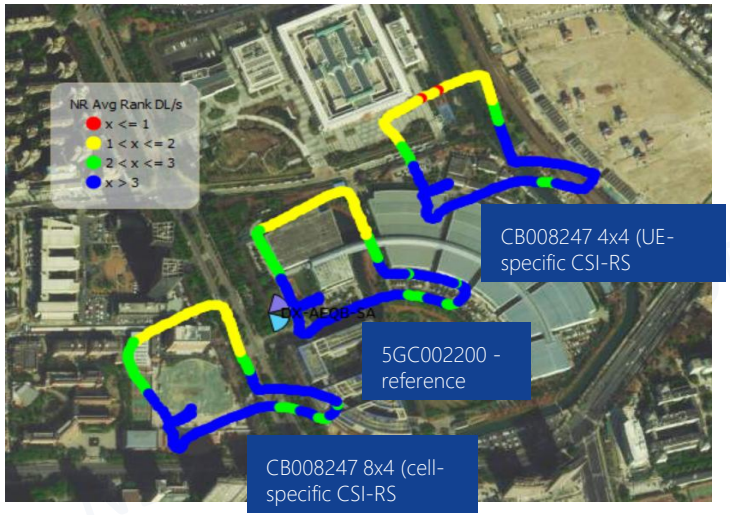
Nokia internal use

Observed Performance Results

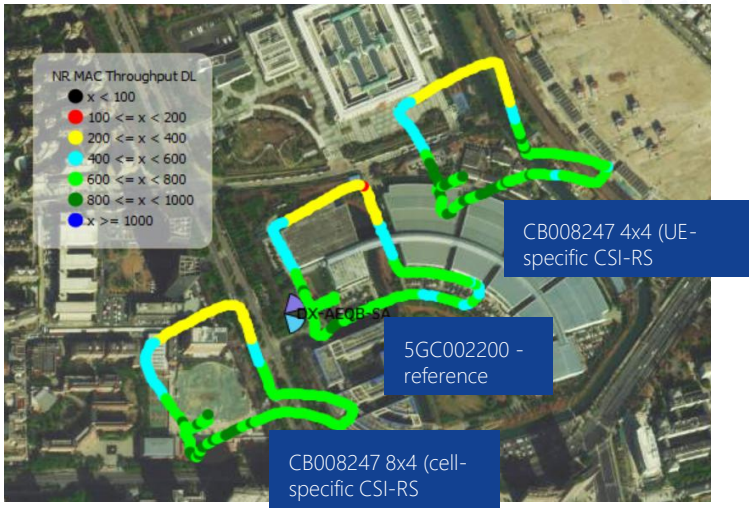
CIS SSD EAST field results

Source: https://nokia.sharepoint.com/:p:/r/sites/MNRITSSDEAST/Shared%20Documents/50%20Biz%20Program/CIS_SSD_East1_FTN/07.PoC_Demo_TF/8247/CB8247%20Option3.2%20result%20.pptx?d=w3ac215c57bee471b9dc9a322ea6b2504&csf=1&web=1&e=qelmD4

NR Rank



NR MAC Throughput DL



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Observed Performance Results

CIS SSD EAST field results

Source: https://nokia.sharepoint.com/:p:/r/sites/MNRITSSDEAST/Shared%20Documents/50%20Biz%20Program/CIS_SSD_East1_FTN/07.PoC_Demo_TF/8247/CB8247%20Option3.2%20result%20.pptx?d=w3ac215c57bee471b9dc9a322ea6b2504&csf=1&web=1&e=qelmD4

Results for all test route

log_file	run	gain	avg_phy_cell_thrpt	NR MAC Throughput DL	NR PDCCH DL GrantCount	NR DL Avg MCS	NR Avg Rank DL/s	PRB Num DL/slot	SS-RSRP	SS-SINR	NR WB CQI	NR PDSCH BLER	256QAM Ratio DL/s	NR PDCP Throughput DL	NR PHY Throughput DL
2200	run1	baseline	574.29	573.03	1218.78	19.40	3.20	269.74	-85.53	31.01	10.89	9.12	45.13	569.41	636.56
	run3			575.55	1218.68	19.30	3.26	269.73	-85.79	31.08	10.95	9.12	43.91	571.84	639.44
8247_4x4	run1	7.62%	618.03	622.98	1224.94	20.46	3.28	269.72	-85.74	30.91	10.53	8.99	62.40	619.00	690.98
	run2			613.08	1224.78	19.90	3.31	269.73	-85.41	31.09	10.38	9.08	54.63	609.11	680.69
8247_8x4	run1	6.08%	609.18	604.90	1218.79	20.36	3.21	269.75	-85.07	31.18	10.84	9.05	57.86	601.03	671.54
	run2			613.47	1218.76	20.77	3.19	269.74	-85.66	30.79	10.78	9.07	65.80	609.56	680.84

Results for rank3&4 area

log_file	run	gain	avg_phy_cell_thrpt	NR MAC Throughput DL	NR PDCCH DL GrantCount	NR DL Avg MCS	NR Avg Rank DL/s	PRB Num DL/slot	SS-RSRP	SS-SINR	NR WB CQI	NR PDSCH BLER	256QAM Ratio DL/s	NR PDCP Throughput DL	NR PHY Throughput DL
2200	run1	baseline	641.44	641.93	1218.80	20.15	3.50	269.74	-81.09	34.60	11.26	9.12	55.53	637.83	712.78
	run3			640.94	1218.74	19.96	3.53	269.74	-81.67	34.38	11.26	9.12	52.67	636.83	711.88
8247_4x4	run1	8.62%	696.75	705.52	1225.07	21.27	3.62	269.72	-81.39	34.36	11.48	8.96	73.51	701.06	782.12
	run2			687.98	1224.78	20.72	3.63	269.72	-81.09	34.45	11.33	9.07	66.73	683.56	763.47
8247_8x4	run1	7.05%	686.65	682.22	1218.77	21.25	3.52	269.74	-80.28	34.86	11.17	9.03	70.28	677.83	756.93
	run2			691.09	1218.80	21.76	3.49	269.73	-81.08	34.52	11.14	9.06	79.12	686.72	766.66

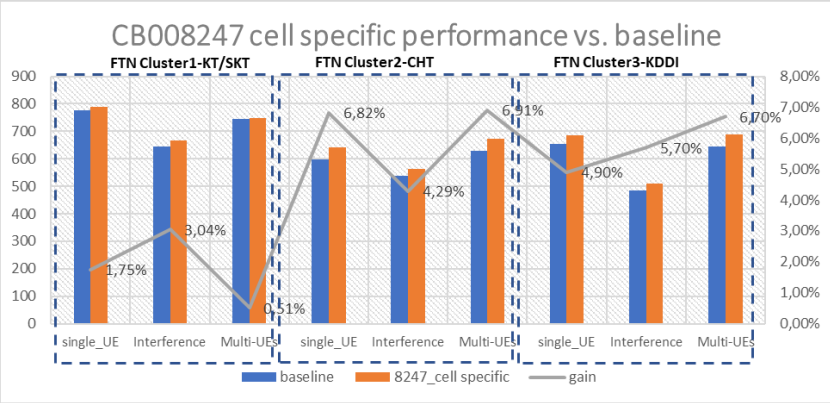
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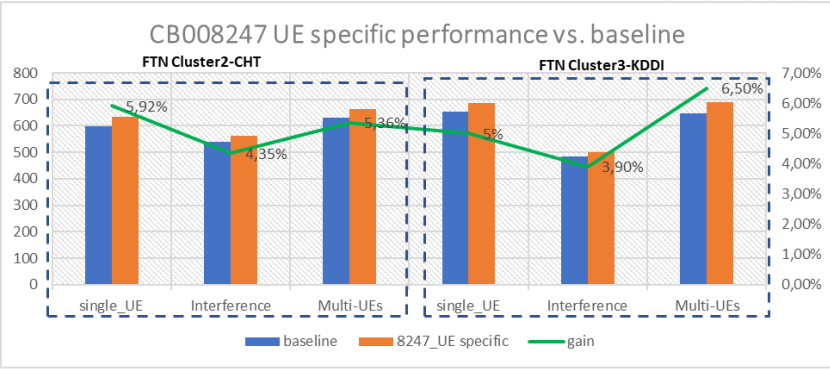
Nokia internal use

Observed Performance Results

CIS SSD EAST field results

Source: https://nokia.sharepoint.com/:p:/r/sites/MNRITSSDEAST/Shared%20Documents/50%20Biz%20Program/CIS_SSD_East1_FTN/06.CIS%20FTN%20feature%20test/8.%20CIS%2022R3%20FTN%20feature%20test/22R3%20performance%20feature%20report%20summary/CIS%20SSD%20East1%20FTN%2022R3%20Performance%20feat





RRM expected feature gain: Avg DL Tput gain:4.5%-9%

Highlight

- SKT/KT used Henrik Beams as the baseline, details as below:
  - DL: **0.51%~3.04%** gain observed on AEQN config, details in link
- KDDI and CHT used the 5GC2200 default beams as baseline, details as below
  - DL: **4.2%~6.9%** Gain on CHT AEQZ config, details in Link
  - DL: **4.9%~6.7%** Gain on KDDI AEQV config, details in Link;
  - Cell specific CIS_RS and UE specific CIS RS config has similar performance Gain

Test environment

FTN Cluster1-KT/SKT

- NR site : ASIB+ABIO+AEQN*3 N78/100M (**2SSB**)
- Traffic model : DL TCP, DVT test
- UE : S22(X65)/S20 (X55)

FTN Cluster2-CHT

- NR site : ASIB+ABIO+AEQZ*2 N78/90M
- Traffic model : DL TCP, DVT test
- UE : S22(X65)

FTN Cluster5-KDDI

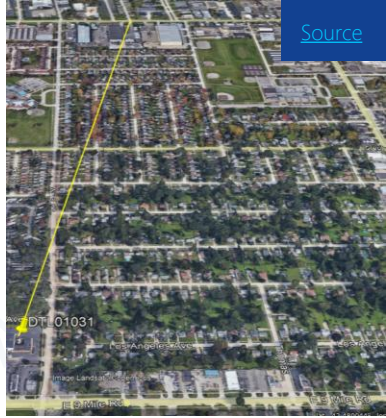
- NR site : ASIB+ABIO+AEQV*3 N77/100M
- Traffic model : DL TCP, DVT test
- UE : Xiaomi11 (X60)

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Nokia internal use

Observed Performance Results

AT&T Dallas FTN CB008247 DL SRS DFT Beam Drive Test Summary



These test runs were completed with SW 22R3 0.1PD (SBTS22R3_ENB_0000_000317_000000)

Radio: AEQK (64TRX)
Beamset: 5+1 120 deg
DFT SRS Config: 14 x 29 (643 unique beams)
Downtilt: -2 deg
Height (m): 31.4
UE: MTP SDX65
actUeCsirsCoexistCellCsirs: true

Conclusions from **CB008247 field** testing on DTFN001031:

- Gains are present in Avg DL SE, MCS, and TxL generally **across entire RSRP range**
- Gains **beat expectations** for CB008247 SRS DFT Beams over 2200 DL PMI beams

Item	06102022 Alpha DL OFF	06092022 8247 ON	Gain 1
Avg RSRP (dBm)	-99.6	-95.7	-4.0%
Avg DL MCS	14.0	17.9	28.1%
Avg DL BLER	12.0	12.1	0.3%
Avg DL MAC TP	145.9	221.2	51.6%
Peak DL MAC TP	329.6	389.2	18.1%
10%-tile DL MAC TP	35.1	127.3	262.4%
90%-tile DL MAC TP	274.8	357.1	29.9%
Avg SE	11.5	17.5	51.5%
Avg RBs	100.8	101.5	0.7%
Avg slot util	66.2	68.6	3.6%
Avg TxL	2.2	2.7	21.8%
Geo-bins	102.0	64.0	-37.3%

Item	06102022 Beta DL OFF	06092022 8247 ON	Gain 1
Avg RSRP (dBm)	-93.9	-96.0	2.2%
Avg DL MCS	14.7	15.6	6.1%
Avg DL BLER	8.9	8.6	-4.1%
Avg DL MAC TP	154.7	177.8	14.9%
Peak DL MAC TP	327.3	383.9	17.3%
10%-tile DL MAC TP	44.1	21.6	-51.0%
90%-tile DL MAC TP	283.8	327.6	15.4%
Avg SE	12.1	14.0	15.9%
Avg RBs	101.6	100.5	-1.1%
Avg slot util	69.6	67.6	-2.8%
Avg TxL	2.2	2.3	5.9%
Geo-bins	134.0	134.0	0.0%

Item	06102022 Gamma DL OFF	06092022 8247 ON	Gain 1
Avg RSRP (dBm)	-95.0	-94.9	-0.1%
Avg DL MCS	15.4	18.4	19.8%
Avg DL BLER	9.0	12.1	33.8%
Avg DL MAC TP	172.0	220.4	28.1%
Peak DL MAC TP	322.6	446.8	38.5%
10%-tile DL MAC TP	63.3	90.0	42.2%
90%-tile DL MAC TP	282.4	352.4	24.8%
Avg SE	13.6	17.3	27.0%
Avg RBs	101.1	101.6	0.5%
Avg slot util	67.5	68.9	2.2%
Avg TxL	2.4	2.6	9.5%
Geo-bins	80.0	80.0	0.0%

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Nokia internal use

KPIs & Counters

CB008247 feature
doesn't introduce new KPIs and Counters



Presented gains shall not be deemed as guaranteed - final figures will always depend on several static or dynamic conditions
Nokia internal use

PM measurements, counters & KPIs

Top/Scorecard KPIs used by QoE for NPIs

- Feature does not introduce any new counters/KPIs
- Following legacy counters can be used to monitor feature impact and performance
  - DL cell/user throughput
  - DL MCS
  - DL CQI

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Nokia internal use

Verification of performance expectations

How to measure and proof feature's impact on performance

Feature impact	How to measure?
DL throughput: improvement After feature activation both cell peak and average throughput may be increased. Cell peak throughput increase will be visible when CSI multiplexing is activated	KPIs: Cell level: • **NR_1481** 5G DL PDCP SDU throughput on NR leg per DRB for QoS group 5 to 20 • **NR_1483** 5G Maximum DL PDCP SDU end user throughput per DRB (note: includes LTE TP) *Note: above KPIs need to be activated with* NRBTS.actIpThpt *parameter* • **NR_5258a** 5G Maximum MAC SDU Cell Throughput in DL on DTCH BTS level: • **NR_5094a** 5G SA PDCP SDU throughput in downlink • **NR_5084a** 5G NSA PDCP SDU throughput (without repetitions) in DL

Presented gains shall not be deemed as guaranteed - final figures will always depend on several static or dynamic conditions

Nokia internal use

Verification of performance expectations

How to measure and proof feature's impact on performance

Feature impact	How to measure?
DL MCS: improvement **DL spectral efficiency:** improvement increase caused by the 2D DFT beamset should result in higher order MCS selected for DL transmission. In case of increase Rank 3/4 usage, this effect may go unnoticed or there may even be a decrease in average MCS (DL throughput should still be improved though).	KPIs: **NR_5068a** 5G Average MCS used in downlink for PDSCH with 256QAM table **NR_5108c** 5G Spectral efficiency in downlink

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Nokia internal use

Verification of performance expectations

How to measure and proof feature's impact on performance

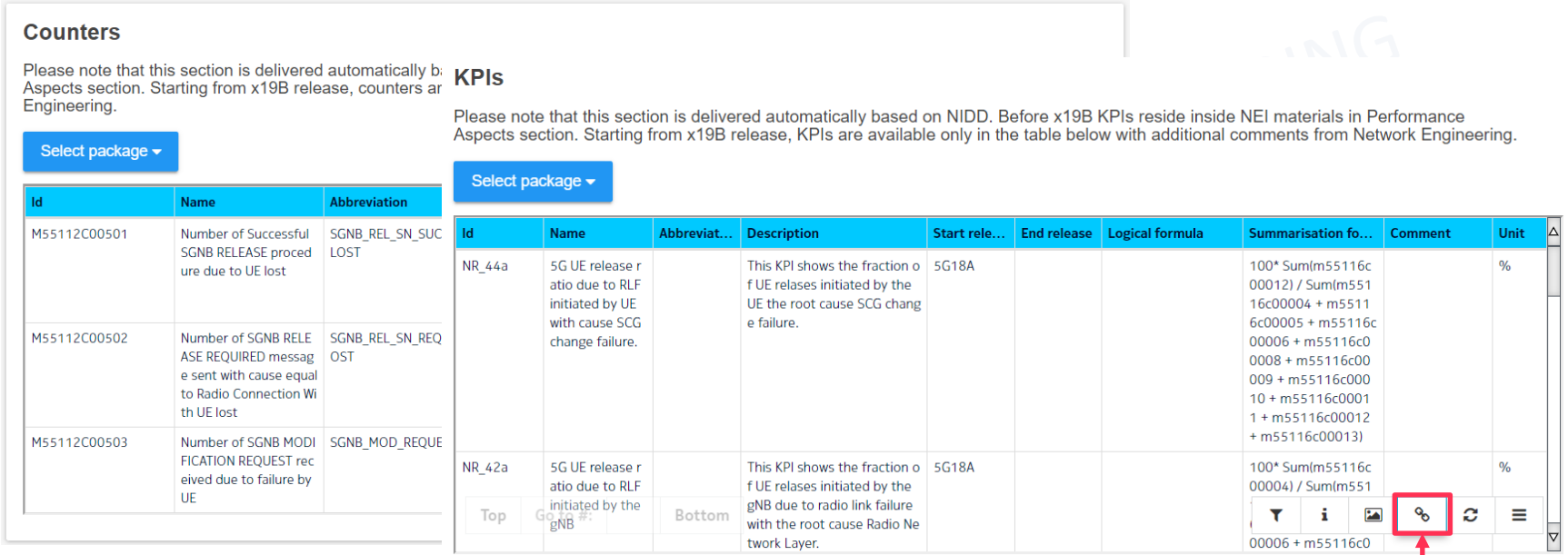
Feature impact	How to measure?
DL CQI: Improvement or no gain In case of **UE specific CSI-RS** being configured, CSI-RS used for CQI estimation are **beamformed** with SRS beam, gain can be observed; In case of **cell-specific CSI-RS**, CQI is internally compensated by SRS beamforming gain difference but **not reflected** by the counters: no gain can be observed	KPIs: **NR_5061b** 5G Average wideband CQI, 256QAM table

Presented gains shall not be deemed as guaranteed - final figures will always depend on several static or dynamic conditions

Nokia internal use

How to find feature related counters & KPIs in WebNEI?

- Under feature materials in WebNEI platform there are dedicated sections called Counters and KPIs, which lists all those related to given functionality



Counters

Please note that this section is delivered automatically b
Aspects section. Starting from x19B release, counters ar
Engineering.

Select package

Id	Name	Abbreviation
M55112C00501	Number of Successful SGNB RELEASE proced ure due to UE lost	SGNB_REL_SN_SUC LOST
M55112C00502	Number of SGNB RELE ASE REQUIRED messag e sent with cause equal to Radio Connection Wi th UE lost	SGNB_REL_SN_REQ OST
M55112C00503	Number of SGNB MODI FICATION REQUEST rec eived due to failure by UE	SGNB_MOD_REQUE

KPIs

Please note that this section is delivered automatically based on NIDD. Before x19B KPIs reside inside NEI materials in Performance Aspects section. Starting from x19B release, KPIs are available only in the table below with additional comments from Network Engineering.

Select package

Id	Name	Abbreviat...	Description	Start rele...	End release	Logical formula	Summarisation fo...	Comment	Unit
NR_44a	5G UE release r atio due to RLF initiated by UE with cause SCG change failure.		This KPI shows the fraction o f UE relases initiated by the UE the root cause SCG chang e failure.	5G18A			100* Sum(m55116c 00012) / Sum(m551 16c00004 + m5511 6c00005 + m55116c 00006 + m55116c0 0008 + m55116c00 009 + m55116c000 10 + m55116c0001 1 + m55116c00012 + m55116c00013)		%
NR_42a	5G UE release r atio due to RLF initiated by the gNB		This KPI shows the fraction o f UE relases initiated by the gNB due to radio link failure with the root cause Radio Ne twork Layer.	5G18A			100* Sum(m55116c 00004) / Sum(m551 00006 + m55116c0		%

- If nesesery you can always download a copy of the KPIs/counters by pushing a button in right corner

Presented gains shall not be deemed as guaranteed - final figures will always depend on several static or dynamic conditions

Nokia internal use

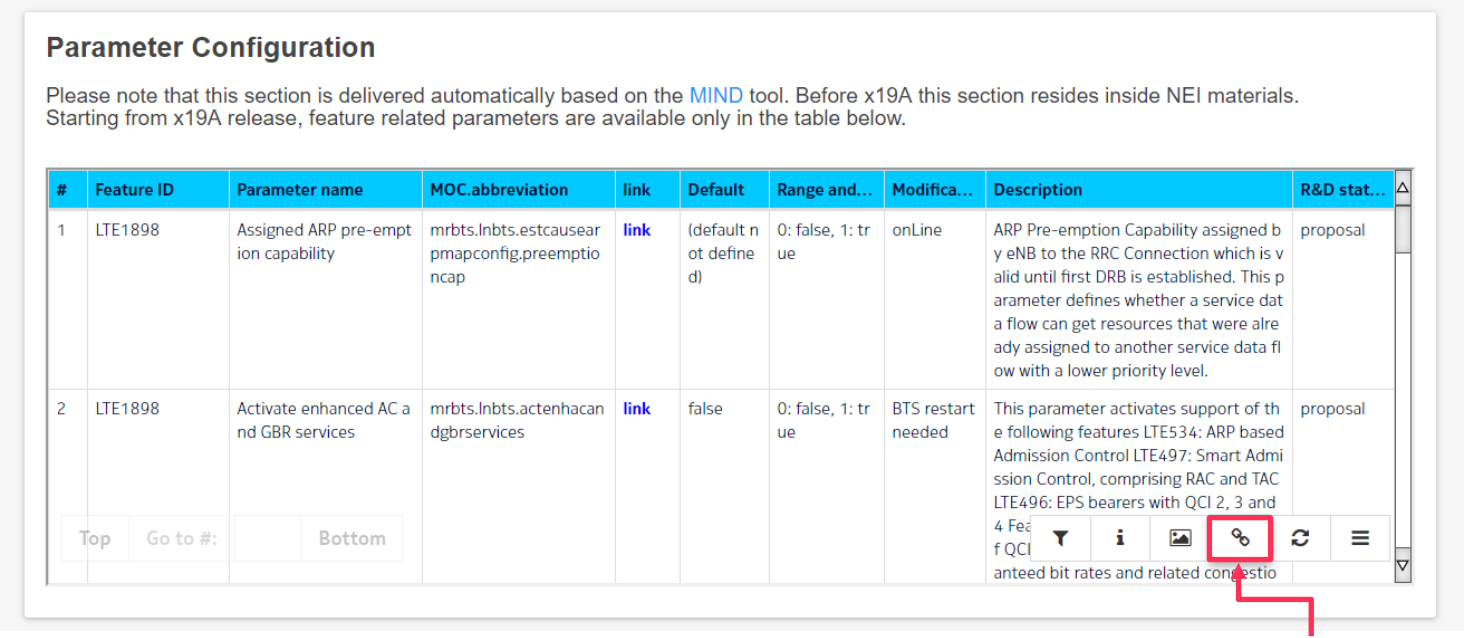
Feature Configuration & Performance Evaluation



Presented gains shall not be deemed as guaranteed - final figures will always depend on several static or dynamic conditions
Nokia internal use

How to find feature parameters in WebNEI?

- Under feature materials in WebNEI platform there is dedicated section called Parameter Configuration, which lists all parameters related to given functionality



Parameter Configuration

Please note that this section is delivered automatically based on the MIND tool. Before x19A this section resides inside NEI materials. Starting from x19A release, feature related parameters are available only in the table below.

#	Feature ID	Parameter name	MOC.abbreviation	link	Default	Range and...	Modifica...	Description	R&D stat...
1	LTE1898	Assigned ARP pre-emption capability	mrbts.lnbts.estcausearpmapconfig.preemptioncap	link	(default not defined)	0: false, 1: true	onLine	ARP Pre-emption Capability assigned by eNB to the RRC Connection which is valid until first DRB is established. This parameter defines whether a service data flow can get resources that were already assigned to another service data flow with a lower priority level.	proposal
2	LTE1898	Activate enhanced AC and GBR services	mrbts.lnbts.actenhacandgbrservices	link	false	0: false, 1: true	BTS restart needed	This parameter activates support of the following features LTE534: ARP based Admission Control LTE497: Smart Admission Control, comprising RAC and TAC LTE496: EPS bearers with QCI 2, 3 and 4 Fea f QCI anteed bit rates and related congestio	proposal

Top | Go to #: | Bottom

- If necessary, you can always download a copy of the parameters by pushing a button in right corner

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Nokia internal use

Feature configuration
Configuration Guides

Configuration Guides by A&S (more about Config Guide concept) define the recommended parametrization for a BTS functionality or supported BTS configurations

There is no separate configuration guide dedicated to 5GC001086 exclusively. Following configuration guideline is related to 5GC001086:

Split mode configuration on mMIMO Radios

Split mode is where antenna array is divided by **horizontally** or **vertically** i.e. top/bottom, left/right sub-panel mode based on radio capability.
Radio can be configured as full TRX mode, full split mode or half split mode
Main use case for split mode config include RF sharing withing same technology NR+NR or across different technology NR+LTE.

Only up/down split **supported** with 5GC001086 on 64/32 TRX RUs

Supported settigs of NRCELL.mMimoAntArrayMode:

Value	Name
0	Full 64TRX Array (4x8x2)
~~1~~	~~Left 32TRX Array (4x4x2)~~
~~2~~	~~Right 32TRX Array (4x4x2)~~
3	Top 32TRX Array (2x8x2)
4	Bottom 32TRX Array (2x8x2)
5	Top 16TRX Array (1x8x2)
6	Bottom 16TRX Array (1x8x2)
7	Full 32TRX Array (2x8x2)

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Nokia internal use

Feature Configuration & Performance Evaluation

Prerequisites & feature activation

Configuration with 6 TDD cells eCPRI 8DL4UL

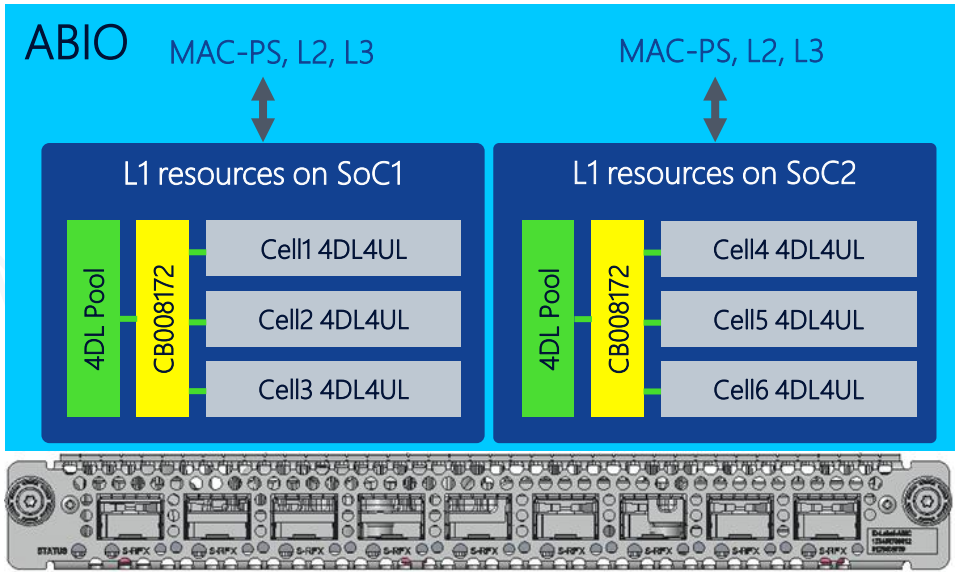
CB006559 NR ABIO BB cell sets up to 6 TDD cells eCPRI 8DL/4UL feature introduces several baseband cell set with up to 3 cells NR FR1 TDD 8DL layers on half ABIO, with eCPRI fronthaul. ABIO unit can support two baseband cell sets - on ABIO slot-A and slot-B (both halves).

CB008172 Baseband pooling of up to 8 DL MIMO layers - FR1 TDD ABIO feature introduces the pooling functionality for FR1 TDD deployments, both SA and NSA on ABIO.

BB capacity pooling is active when parameter MRBTS.NRBTS.actSpatialStreamPooling is set to true and a given partition of gNB-DU is configured with multiple L1 spatial stream that belong to the same BB pool.

4 DL MIMO layers are pooled across the cells of one ABIO slot. 1 cell out 3 have peak 8DL4UL capacity, while remaining 2 cells have 4DL4UL capacity.

If there are less then 3 cells in the set, no polling is necessary, since there are enough L1 resources to support dedicated 8DL/4UL processing.



More details: CB008172 webNEI

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Feature Configuration & Performance Evaluation

Prerequisites & feature activation

Prerequisites

- Cell, which should be enabled with CB008247 must be set with TDD technology – **NRCELL.cellTechnology** should be set to **'1'** (TDD).
- 5GC001086 must be enabled: feature activation flag (NRCELL. actulSrsBm = true)
- Feature activation flag (NRCELL. actDlSrsBm = true)
- Beamforming must be enabled (NRCELL.actBeamforming = true)
- Feature is supported **only on ABIO**
- Feature is applicabel to **eCPRI only**
- Feature is applicable ot long PUCCH only (Parameter NRCELL.pucchModeSelect should be set to the value '1')
- Paremeter NRCELL.mMimoAntArrayMode needs to be **defined**
- RU must be **32TRX or 64TRX** (see 5GC001086 webNEI for detailed list)
  - **Until 23R3** for 64TRX RUs, CB008247 works with 8DL4UL configurations (nrCellType=8DL4UL). With **CB010388** in **23R3**, CB008247 **works with both 8DL4UL and 4DL4UL** configurations
  - For 32TRX RUs, nrCellType= 8DL4UL and 4DL4UL **are supported**

⚠ CB008247 feature **cannot be enabled together with** legacy 5GC000524 DL MU-MIMO feature.

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Feature Configuration & Performance Evaluation

Prerequisites & feature activation

Activation Recommendations

- It is recommended to activate CB008247 together with 5GC001086
  - Activation of 5GC001086 allows for the UEs that do not support TAS SRS to use codebook SRS with DL beamforming
- Updated cSRS and bSRS parameter recommendations – see Technical Details

- In some cases *actUeCsirsCoexistCellCsirs=true* parameter setting may lead to performance decrease compared to *actUeCsirsCoexistCellCsirs=false* due to fallback to SSB beams. What can be tried:
  - Setting *actUeCsirsCoexistCellCsirs=false*
  - Decreasing *srsPowerThresholdForDlBm* (available from 23R1), but it may lead to increased BLER
  - Improving SSB performance

- In 22R4 parameter *NRCELL.tmpParam3* needs to be set to recommended value 1200 corresponding to -120dBm. This parameter is responsible for defining the SRS received power threshold for fallback to legacy beamforming

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Feature Configuration & Performance Evaluation

Application area selection

Feature **is expected** to provide gain in DL cell and user throughput **regardless** of the existing radio and load conditions.

Actual feature gain comes from the application of the advanced and denser 2D DFT beamset as well as best beam selection.

Simulations indicate that the gain should be visible for both single UE and multiple UE scenarios.

Performance optimization TF is ongoing.
Remaining performance optimization work will be done under 23R1.

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Configuration Management <CB008247>

New parameters

Online Modifable | Requires Object Locking | Requires BTS Restart

MOC.{Structure}.Abbreviated name	Full name/Description	Range and Step		Unit	Default
NRCELL. actDlSrsBm	Activate downlink SRS based beam management This parameter activates/deactivates downlink SRS based beam management. Updating this parameter will provoke a cell reset.	Value	Name	[-]	0
		1	true		
		0	false		

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Nokia internal use

Configuration Management <CB008247>

New parameters

Online Modifable | Requires Object Locking | Requires BTS Restart

MOC.{Structure}.Abbreviated name	Full name/Description	Range and Step		Unit	Default
NRCELL. actCsirsMultiplexing	Activate multiple CSI-RS resources multiplexing in slot Activation flag for multiple CSI-RS resources multiplexing in slot This parameter activates/deactivates multiple CSI-RS resources multiplexing in slot for both UE and cell specific CSI-RS.	Value	Name	[-]	0
		1	true		
		0	false		

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Configuration Management <CB008247>

New parameters

Online Modifable | Requires Object Locking | Requires BTS Restart

MOC.{Structure}.Abbreviated name	Full name/Description	Range and Step		Unit	Default
NRCELL.actUeCsirsCoexistCellCsirs	Activate UE and cell specific CSI-RS coexist Activation flag for UE specific CSI-RS coexisted with cell specific CSI-RS in case NRCELL.actDlSrsBm is equal to TRUE and NRCELL.dlMimoMode is equal to '8x4 Closed Loop Spatial Multiplexing' Updating this parameter will provoke a cell reset. FALSE means no UE specific CSI-RS resources are allocated by the gNB. TRUE means the gNB will firstly allocate UE specific CSI-RS resources.	Value 1 0	Name true false	[-]	0

If NRCELL.actDlSrsBm is equal to TRUE and NRCELL.dlMimoMode is equal to '8x4 Closed Loop Spatial Multiplexing', then NRCELL.actUeCsirsCoexistCellCsirs shall be configured.

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Configuration Management <CB008247>

New parameters

Online Modifable | Requires Object Locking | Requires BTS Restart

MOC.{Structure}.Abbreviated name	Full name/Description	Range and Step		Unit	Default
NRCELL. srsForBeamMgmt.bSRS	SRS subband configuration This parameter indicates the SRS subband configuration (corresponding to BSRS in table 6.4.1.4.3-1 of 3GPP TS38.211). Together with NRCELL.srsForBeamMgmt.cSRS, it defines the subband bandwidth of periodic SRS which is configured for a UE for both codebook SRS and antenna switching SRS. Please refer to table 6.4.1.4.3-1 in 3GPP TS38.211 for different system bandwidth.	Min	0	[-]	0
		Max	3		
		Step	-		

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Configuration Management <CB008247>

New parameters

Online Modifable | Requires Object Locking | Requires BTS Restart

MOC.{Structure}.Abbreviated name	Full name/Description	Range and Step	Unit	Default
NRCELL. srsForBeamMgmt	SRS for beam management This structure gathers parameters for SRS based beam management.	[-]	[-]	0

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Configuration Management <CB008247>

New parameters

Online Modifable | Requires Object Locking | Requires BTS Restart

MOC.{Structure}.Abbreviated name	Full name/Description	Range and Step		Unit	Default
NRCELL. srsForBeamMgmt.cSRS	SRS bandwidth configuration This parameter indicates the SRS bandwidth configuration (corresponding to CSRS in table 6.4.1.4.3-1 of 3GPP TS38.211). Together with NRCELL.srsForBeamMgmt.bSRS, it defines the subband bandwidth of the periodic SRS which is configured for a UE for both codebook SRS and antenna switching SRS. Please refer to table 6.4.1.4.3-1 in 3GPP TS38.211 for different system bandwidth.	Min	0	[-]	0
		Max	63		
		Step	-		

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Configuration Management <CB008247>

Online Modifable | Requires Object Locking | Requires BTS Restart

New parameters

MOC.{Structure}.Abbreviated name	Full name/Description	Value	Name	Unit	Default
NRCELL. srsForBeamMgmt. srsTypeAllocForDlBm	Type of SRS used for SRS based DL beamforming his parameter indicates which type of SRS is used for SRS based DL beamforming. The value 'codebook' means that only codebook SRS is used for DL beam management. The value 'antenna switching' means that only antenna switching SRS is used for DL beam management. The value 'antenna switching or codebook' means that, if UE is capable of antenna switching SRS, then antenna switching SRS is used for DL beam management; otherwise codebook SRS is used for DL beam management. Note: This parameter is applied only if both SRS based DL beamforming and SRS based UL beamforming are activated. If SRS based UL beamforming is deactivated, then no codebook SRS is allocated no matter which parameter value is set.	0	antenna switching	[-]	2
		1	codebook		
		2	antenna switching or codebook		

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Technical Details

Nokia product specific SRS bandwidth configurations

1st candidate								2nd candidate					
Channel Bandwidth (chBw)	PRB amount	SRS Bandwidth (cSRS)	SRS Subband (bSRS)	SRS frequency range	Subband size	SRS Capacity per slot	SRS vs BW	SRS Bandwidth (cSRS)	SRS Subband (bSRS)	SRS frequency range	Subband size	SRS Capacity per slot	SRS vs BW
20	51	12	2	48	8	48	94.12%						
30	78	19	2	72	12	48	92.31%	20	2	76	4	152	97.44%
40	106	23	2	96	16	48	90.57%	25	2	104	4	208	98.11%
50	133	32	2	128	8	128	96.24%						
60	162	38	2	144	8	144	88.89%	42	2	160	16	80	98.77%
70	189	42	2	160	16	80	84.66%	45	2	184	4	368	97.35%
80	217	49	2	192	8	192	88.48%	NOTE: SRS subband size 8-16 PRB is the first recommendation. 4 PRB works fine in simulation and is available for field trials. Smaller SRS subband size provides higher SRS capacity and wider coverage. 4 PRB can be used if it works fine in field after testing.					
90	245	55	3	240	8	240	97.96%						
100	273	63	2	272	8	272	99.63%						

- The defined values represent balance between the SRS capacity and performance.
- If no specific requirement e.g. single UE peak throughput test, then 1st candidate should be used as default settings.

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Configuration Management <CB008247>

New parameters

Online Modifable | Requires Object Locking | Requires BTS Restart

MOC.{Structure}.Abbreviated name	Full name/Description	Range and Step		Unit	Default
NRCELL. numDftBeamMuteLeft	Number of leftmost DFT beams muted Number of leftmost DFT beams in horizontal which are muted. This parameter indicates the number of leftmost DFT beams in horizontal which are muted. The number of beams muted is relative to the full DFT beam set. Value 0 means the full DFT beam set in left side is in use. Value 1 means 1 beam relative to the full DFT beam set in left side is muted. And so on... If this parameter is not configured, then the DFT beams in left side are internally determined by gNB according to the open angle setting.	Min	0	[-]	[-]
		Max	16		
		Step	-		
NRCELL. numDftBeamMuteRight	Number of rightmost DFT beams muted Number of rightmost DFT beams in horizontal which are muted. This parameter indicates the number of rightmost DFT beams in horizontal which are muted. The number of beams muted is relative to the full DFT beam set. Value 0 means the full DFT beam set in right side is in use. Value 1 means 1 beam relative to the full DFT beam set in right side is muted. And so on... If this parameter is not configured, then the DFT beams in right side are internally determined by gNB according to the open angle setting.	Min	0	[-]	[-]
		Max	16		
		Step	-		

Nokia internal use

Configuration Management <CNI-92005>

New parameters

Online Modifable | Requires Object Locking | Requires BTS Restart

MOC.{Structure}.Abbreviated name	Full name/Description	Range and Step		Unit	Default
NRCELL. muteLastSymbolInSsbSlot	Mute last symbol in SSB slot This parameter enables/disables muting of the last symbol in SSB slot(s). The possible values are as follows: - 'NoMutingOfDlSymbol' meaning no muting of the last DL symbol in the SSB slots. - 'MutingOfLastDlSymbolInSsbSlotZero' meaning muting of the last DL symbol only in SSB slot 0 (first SSB slot), when no CSI-RS is allocated in the last symbol of the 0th SSB slot. - 'MutingOfLastDlSymbolInAllSsbSlots' meaning muting of the last DL symbol in all SSB slots, when no CSI-RS is allocated in the last symbol of the SSB slot.	Value	Name	[-]	0
		0	NoMutingOfDlSymbol		
		1	MutingOfLastDlSymbolInSsbSlotZero		
		2	MutingOfLastDlSymbolInAllSsbSlots		

CNI-92005: No PDSCH scheduling in last symbol of SSB slot to mitigate high BLER

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Configuration Mangement <CB008247>

Related parameters

Online Modifable | Requires Object Locking | Requires BTS Restart

MOC.{Structure}.Abbreviated name	Full name/Description	Range and Step		Unit	Default
		Value	Name		
NRCELL. actUlSrsBm	Activate uplink SRS based beam management This parameter activates/deactivates uplink SRS based beam management. Updating this parameter will provoke a cell reset.	1	true	[-]	0
		0	false		

It is recommended to enable 5GC001086 feature to support fallback to codebook SRS for UEs not supporting TAS SRS.

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Configuration Management <CB008247>

Online Modifable | Requires Object Locking | Requires BTS Restart

Related parameters

MOC.{Structure}.Abbreviated name	Full name/Description	Range and Step		Unit	Default
NRCELL. beamSet.lowerEdgeAngle	Lower cell edge angle This parameter defines the angle limiting cell on lower side with respect to bore sight. The parameter is applicable and used only in analog beamforming.	Min	45	[degree]	135
		Max	135		
		Step	-		

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Configuration Management <CB008247>

Online Modifable | Requires Object Locking | Requires BTS Restart

Related parameters

MOC.{Structure}.Abbreviated name	Full name/Description	Range and Step		Unit	Default
NRCELL. beamSet.upperEdgeAngle	Upper cell edge angle This parameter defines the angle limiting cell on upper side with respect to bore sight. The parameter is applicable and used only in analog beamforming.	Min	45	[degree]	80
		Max	135		
		Step	-		

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Configuration Management <CB008247>

Related parameters

Online Modifable | Requires Object Locking | Requires BTS Restart

MOC.{Structure}.Abbreviated name	Full name/Description	Range and Step		Unit	Default
NRCELL. mMimoAntArrayMode	Massive MIMO Antenna Array Mode This parameter indicates the massive MIMO (mMIMO) antenna array mode on 64 pipes mMIMO MAA.	0	Full 64TRX Array (4x8x2)	[-]	0
		~~1~~	~~Left 32TRX Array (4x4x2)~~		
		~~2~~	~~Right 32TRX Array (4x4x2)~~		
		3	Top 32TRX Array (2x8x2)		
		4	Bottom 32TRX Array (2x8x2)		
		5	Top 16TRX Array (1x8x2)		
		6	Bottom 16TRX Array (1x8x2)		
		7	Full 32TRX Array (2x8x2)		

*Note: left/right antenna split are not supported by 5GC001086 2D DFT beamsets and therefore cannot be used by CB008247

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Configuration Management < CB008247 >

Related parameters

Online Modifable | Requires Object Locking | Requires BTS Restart

MOC.{Structure}.Abbreviated name	Full name/Description	Range and Step		Unit	Default
NRCELL.nrCellType	NR Cell type This parameter allows the operator to specify the number of fronthaul streams of a New Radio Cell. This parameter is also used for identification of Supplemental downlink cell (FR2). Configuring this parameter during cell addition will not provoke gNB reset. Updating this parameter will provoke gNB reset. Removal of this parameter during cell deletion will not provoke gNB reset.	Value	Name	[-]	-
		20	2DL0UL		
		21	2DL1UL		
		22	2DL2UL		
		24	2DL4UL		
		42	4DL2UL		
		44	4DL4UL		
		82	8DL2UL		
		84	8DL4UL		
		88	8DL8UL		
		168	16DL8UL		
		1616	16DL16UL		

nrCellType=4DL4UL for 32TRX RU deployment and nrCellType=8DL4UL for 64TRX RU deployment
With CB010388 in 23R3 nrCellType=4DL4UL for 64TRX RU deployment becomes possible

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Configuration Management < CB008247 >

Related parameters

Online Modifable | Requires Object Locking | Requires BTS Restart

MOC.{Structure}.Abbreviated name	Full name/Description	Range and Step		Unit	Default
NRCELL.dlMimoMode	Downlink MIMO Mode This parameter selects the DL MIMO mode for the PDSCH. NxM MIMO Notation: N = number of CSI-RS antenna ports used for PMI per UE; M = max number of PDSCH layers (max Rank) per UE. Configuring this parameter during cell addition might provoke gNB reset - gNB takes decision of reset necessity. Updating this parameter will provoke gNB reset. Removal of this parameter during cell deletion will not provoke gNB reset.	Value	Name	[-]	30
		30	2x2 Closed Loop Spatial Multiplexing		
		40	4x4 or 4x2 Closed Loop Spatial Multiplexing		
		50	2x2 Open Loop Spatial Multiplexing		
		60	4x4 or 4x2 Open Loop Spatial Multiplexing		
		70	8x4 Closed Loop Spatial Multiplexing		

Settings applicable to CB008247 feature

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Verification of performance expectations

CDIT+D Considerations

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Verification of performance expectations

CDIT+D Considerations (1)

CDIT+D stands for Continuous Delivery Integration Testing and Development
Aim of the program is to deliver early feature SW drops in selected customers' clusters in order to speed up delivery of stable product.

Software drops are scheduled on bi-weekly basis and will start before P7 milestone of the release.

Latest status of the 22R3 PT trunk report can be found here.

CB008247 feature will be delivered in CDIT+D mode in 22R3 for selected customers starting from PT1 (middle March 2022):

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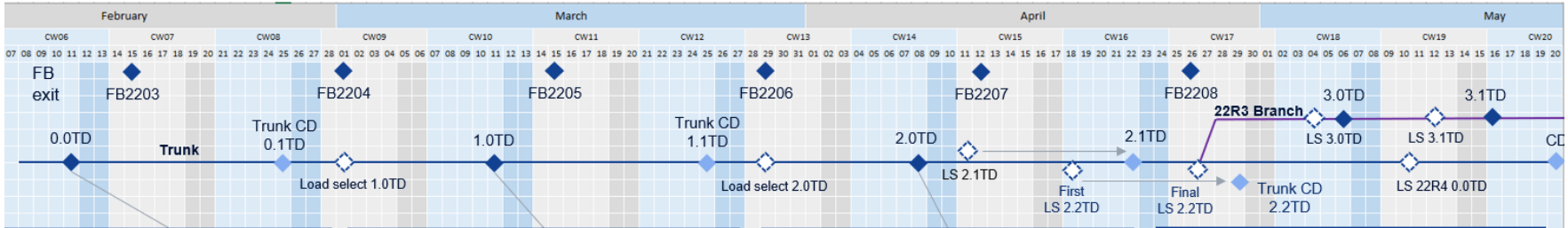
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Verification of performance expectations

CDIT+D Considerations (2)

Source: 22R3 PT trunk report
Status on Mar 15, 2022
See here for most recent status



22R3 0.0TD - TRUNK – RELEASED
Frozen: 1st FEB, Released 11th FEB 2022
Load: SBTS00_ENB_9990_220201_000004
Content:
- Customer TMO US only.
- FB2201, FB2202 feature items
- No TMO required new feature items ready

22R3 1.0TD – TRUNK - RELEASED
Frozen: 1st MAR, Released 11th MAR 2022
Load: SBTS00_ENB_ 9990_220228_000007
Content:
- FB2203, FB2204 feature items "ST done"
- **8247-A** SRS based DL BF (UE specific) – **lab ready**
- **8532-A, 8532-B** 4CC CA (TDD as PCell) – **field ready**
- **8533-A** 4CC CA (FDD as PCell) - **field ready**
- **CNI-70231 - field ready**
- 8697-A UE overheating - not released
- CNI-79164 – not released
- 8379 – not released
- 8132 – not released
- 8128 – not released

22R3 2.0TD – TRUNK - RELEASED
Frozen: 1st APR, Released 8th APR 2022
Load: SBTS00_ENB_ 9990_220328_000005
Content
- 8532-F - Interworking with 5GC002200
- 8532-D Pcell change, Measurement based Scell addition/release (partial)
- 8533-A, 8533-B, 8533-C
- 8697-A – restricted released → 2.1TD
- 8247-A, -B SRS based DL BF (Cell specific) – **limited field trial** (due performance restriction)
- 8128-C#B4, -C#B5, -C#B3 – ST done
  Note: 8128-C#B2 has plan in 3.0TD
- 8132-A (lab as precondition for –B)
- CNI-79164

22R3 2.1TD – TRUNK - RELEASED
Frozen: 15th APR, Release 21st APR 2022
Load: SBTS00_ENB_ 9990_220411_000009
Content
- Corrections on top of 2.0TD

22R3 3.0TD – BRANCH - RELEASED
Freeze: 4th MAY, Release 6th MAY 2022
Load: SBTS22R3_ENB_0000_000092_000000
Content
- **Up to FB2208 released features (ST done)**

CDIT+D content:
- 8532 (full, field), 8533 (full, field)
- 8247-A , -B, -D (DRX) SRS based DL BF - field trial
- 8132-A, -B, -C – field trial
- 8379 (full, field)
- 8697 (full, field)
- 8128 (full, field)
- CNI-75210, CNI-75211 n77 band support AT&T

22R3 3.1TD – BRANCH – RELEASED
Frozen: 12th MAY, Released 16th MAY 2022
Load: SBTS22R3_ENB_0000_000190_000000
Content
- Corrections on top of 3.0TD

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References and acknowledgments

Reference	Version	Author(s)	Link	Date of access
CB008247 CFAM	1.7	Ye Huanqiu		
CB008247 FS1 Technical Analysis	1.0	Li Baocai	Link	
5GC001086 webNEI	1.6	Krzysztof Wisniowski	Link	
22R3 Trunk Report	24-03-2022	Thomas Schwaiger, Gopal Halder	Link	24-03-2022
CB008247system level simulation reports	Mar. 2022	Hu Yuping	link	
5G CFAM Kick-off: CB008247: SRS based DL beamforming	23-06-2021	Ye Huanqiu	link	
Overview of CB008247 – SRS-based DL beamforming	19-06-2021	Vuokko Nurmela	link	

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Backup

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